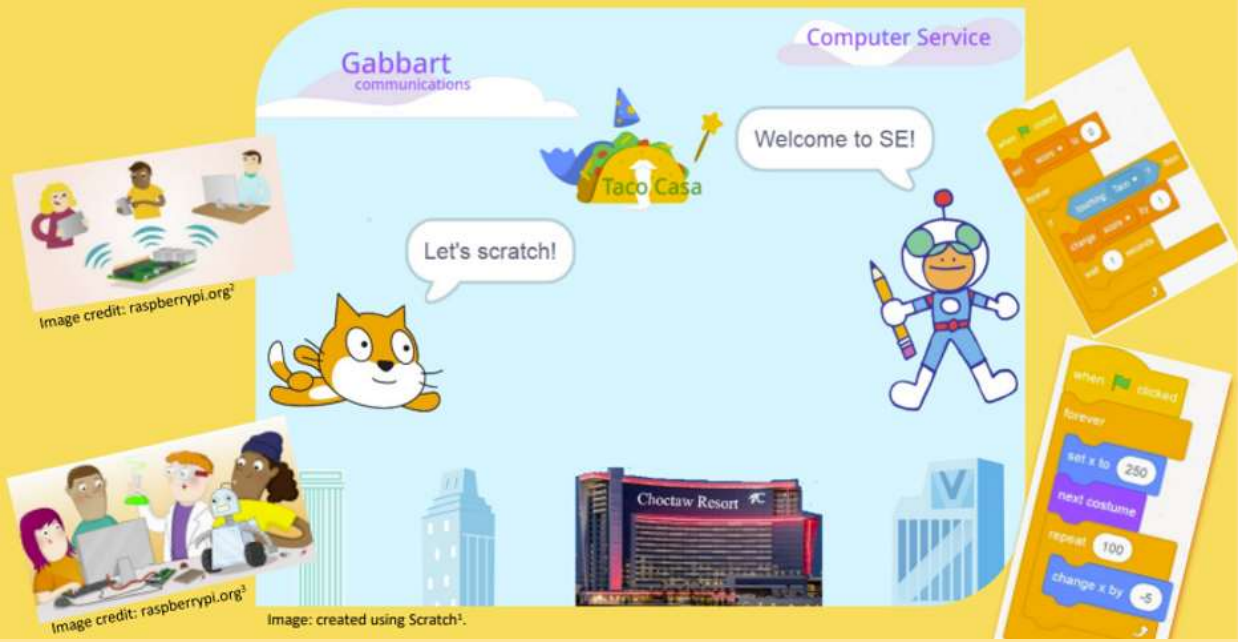
IT-Camp for Kids

IoT (Internet of Things) using Raspberry Pi and Scratch Workshops

Overview



- Dr. Ming-Shan Su
- Computer Science Professor
- Chemistry, Computer and Physical Sciences Department
- Southeastern Oklahoma State University
- IT-Camp Organizer/Instructor

Let's Scratch and Have Fun

Beginner IT Camp

No experience needed!

Image credit: raspberrypi.org[3]

Image credit: raspberrypi.org[3]

Image: created using Scratch[1].

Hands-on IT – STEM[4] Workshops[5,6] - 2022:

IoT (Internet of Things) using Raspberry Pi and Scratch

Workshop Topics:	Workshops (WS)		
Contents* (IoT)	WS-1	WS-2	WS-3
1. Introducing IoT (Internet of Things), Raspberry Pi, and Scratch 3	✓	✓	✓
2. Game Programming - Scratch 3	✓	✓	✓
3. Exploring Scratch Extensions	✓	✓	✓
4. Getting to Know your Raspberry Pi Take a brief guided tour of your new computer	✓	✓	✓
5. Virtual IT Field Trip with Q & A Choctaw Nation – IT Dept.	✓	✓	✓
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10. Programming with Python (will be briefly covered).			✓
11. Let's Stargate the Present and the Future	✓	✓	✓

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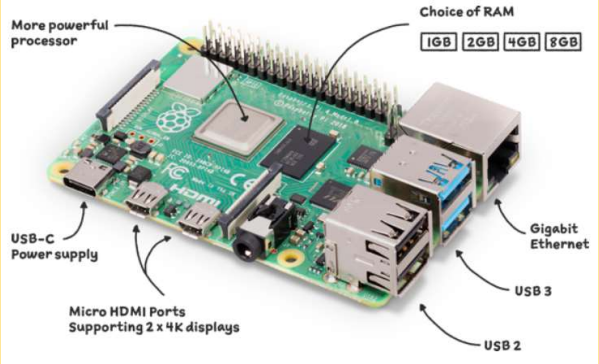
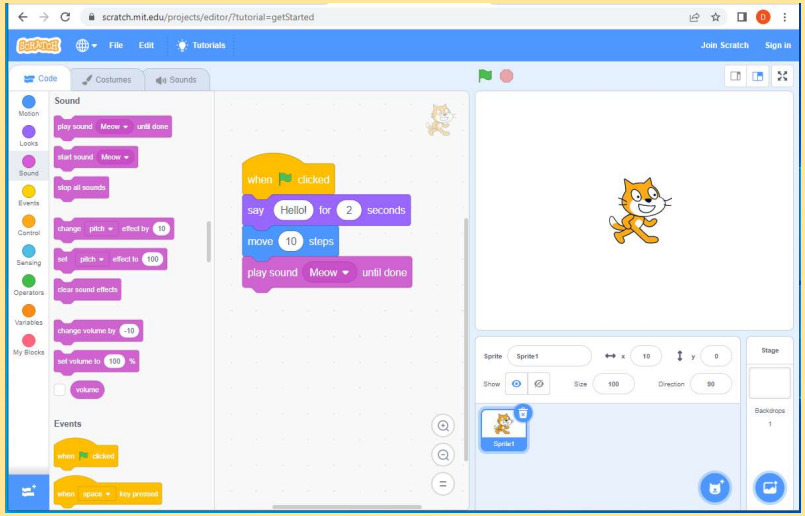
IoT (Internet of Things)

Raspberry Pi



Source: https://www.raspberrypi.com/products/raspberry-pi-4-model-b/

Scratch coding software



Source: https://scratch.mit.edu/

- What is Internet of Things (IoT)?
  - The **Internet of Things** describes physical objects with sensors, processing ability, software, and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks. Wikipedia
- What is a Raspberry Pi?
  - The **Raspberry Pi** is a tiny and affordable computer that you can use to learn programming through fun, practical projects. – https://ww.raspberrypi.org
- What is Scratch?
  - **Scratch** is a free block-based programming language and online community where you can create your own interactive stories, games, and animations. - https://scratch.mit.edu

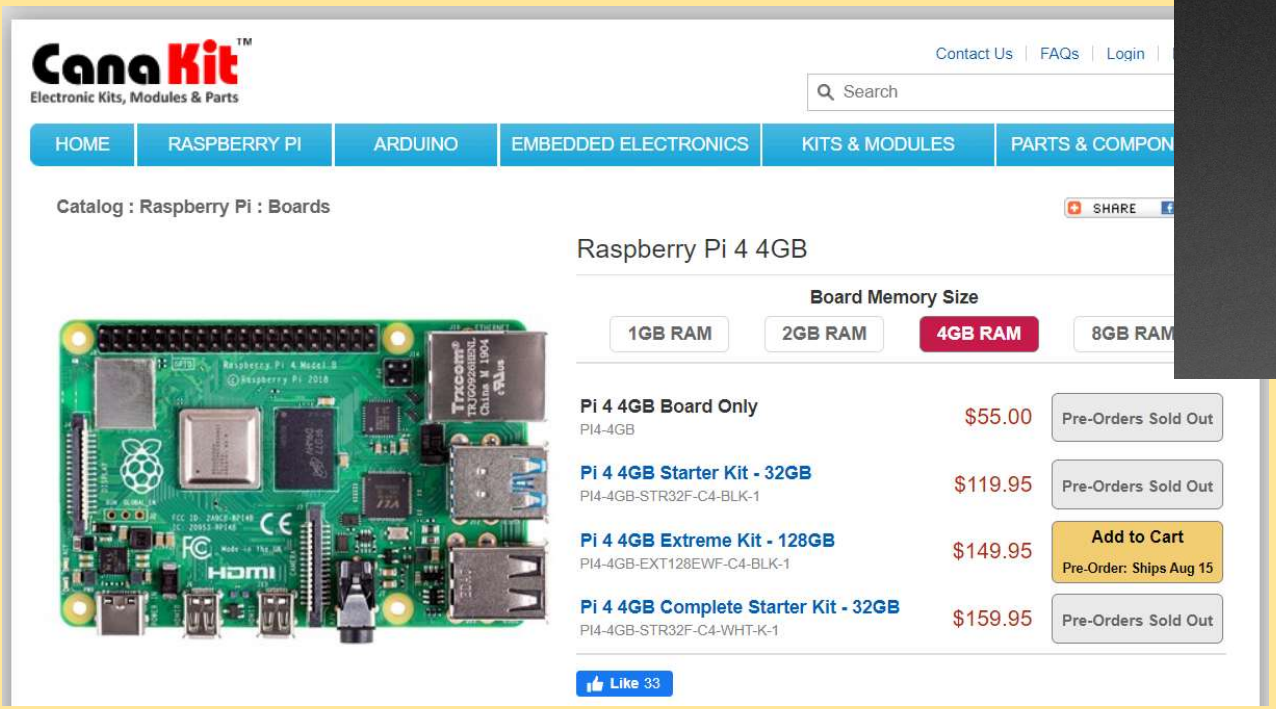
https://youtu.be/LlhmzVL5bm8

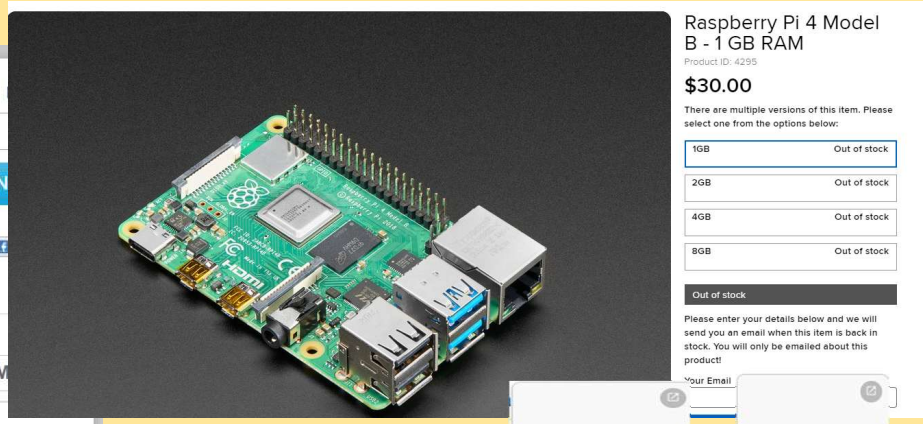
https://youtu.be/uXUjwk2-qx4

https://youtu.be/jXUZaf5D12A

We are so blessed! 😎

- Check the Live website … (Raspberry Pi 4, 4GB)
- https://www.raspberrypi.com/products/raspberry-pi-4-model-b/
- https://www.adafruit.com/product/4295?src=raspberrypi





Raspberry Pi 4 Model B 2019 Quad Core 64 Bit WiFi Bluetooth (1GB)
$125.95
Amazon.com
Free shipping

Raspberry Pi 4 Model B 4 GB RAM 64-bit Quad Core WiFi/BT, RASPBERRY PI...
$99.95
Jameco.com

Global shortage of Chips => Raspberry Pi

Good News: enough Raspberry Pi for our students (:>)

CanaKit - Raspberry Pi 4 Extreme Kit 8GB RAM

Model: PI4-8GB-EXT128EWF-C4-WHT-RT **SKU:** 6429585

(42 reviews)

Sold Out

Compare Save

$169.99

Sold Out

CanaKit - Raspberry Pi 4 Starter PRO Kit 8GB RAM

Model: PI4-8GB-STR32WF-C4-BLK-RT **SKU:** 6429586

(34 reviews)

Sold Out

Compare Save

$139.99

Sold Out

CanaKit - Raspberry Pi 4 Starter PRO Kit 4GB RAM - Premium Black Case

Model: PI4-4GB-STR32W-C4-BLK-RT **SKU:** 6365737

(292 reviews)

Sold Out

Compare Save

$119.99

Sold Out

CanaKit - Raspberry Pi 4 2GB Ultimate Maker Kit - Clear

Model: PI4-2GB-ULT32W-C4-CLR-RT **SKU:** 6365733

(48 reviews)

Sold Out

Compare Save

$119.99

Sold Out

canakit.com/raspberry-pi-4-starter-kit.html

Introducing the new Raspberry Pi 400. Order yours today!

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Catalog : Raspberry Pi : Pi 4 Kits

SHARE

Raspberry Pi 4 Starter Kit

by CanaKit

Pi 4 1GB Starter Kit - 32GB PI4-1GB-STR32F-C4-BLK	$99.95	Add to Cart Pre-Order: Ships Oct 30
Pi 4 2GB Starter Kit - 32GB PI4-2GB-STR32F-C4-BLK	$109.95	Pre-Orders Sold Out
Pi 4 4GB Starter Kit - 32GB PI4-4GB-STR32F-C4-BLK	$119.95	Pre-Orders Sold Out
Pi 4 8GB Starter Kit - 32GB PI4-8GB-STR32F-C4-BLK	$139.95	Pre-Orders Sold Out

Like 43

- Raspberry Pi 4 Model B with 1.5GHz 64-bit quad-core ARMv8 CPU
- 2GB, 4GB or 8GB RAM
- CanaKit 3.5A USB-C Power Supply with Noise Filter (UL Listed) specially designed for the Raspberry Pi 4 (5-foot cable)
- CanaKit Premium Black Case (High Gloss), Set of 3 Aluminum Heat Sinks
- CanaKit Fan
- Micro HDMI to HDMI Cable (6-foot cable)
- SanDisk 32GB Class 10 MicroSD Card, USB Card Reader Dongle
- CanaKit Quick-Start Guide

Product Description

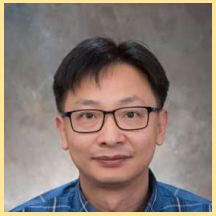
An exclusive Starter Kit for Raspberry Pi from CanaKit that includes the latest generation of the Raspberry Pi family. The kit includes the Raspberry Pi 4 and everything you need to get up and running within minutes in the exciting world of Raspberry Pi!

With the CanaKit brand Raspberry Pi 4 Starter Kit, you can rely on the compatibility, quality, consistency and service that comes with all of the CanaKit brand Raspberry Pi 4 kits.

We are so blessed!

- This IT-Camp would not be possible with the hep and support from the Depart. of CCPS, SE, and the community.

Southeastern Oklahoma State University



IT Support Services

Southeastern Foundation

University Marketing

and so many … others who have helped.

SE Idabel camp

Benjamin Little

Marietta ISD

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scratch.mit.edu

https://youtu.be/jXUZaf5D12A

Let's watch the video.

- What is Scratch?
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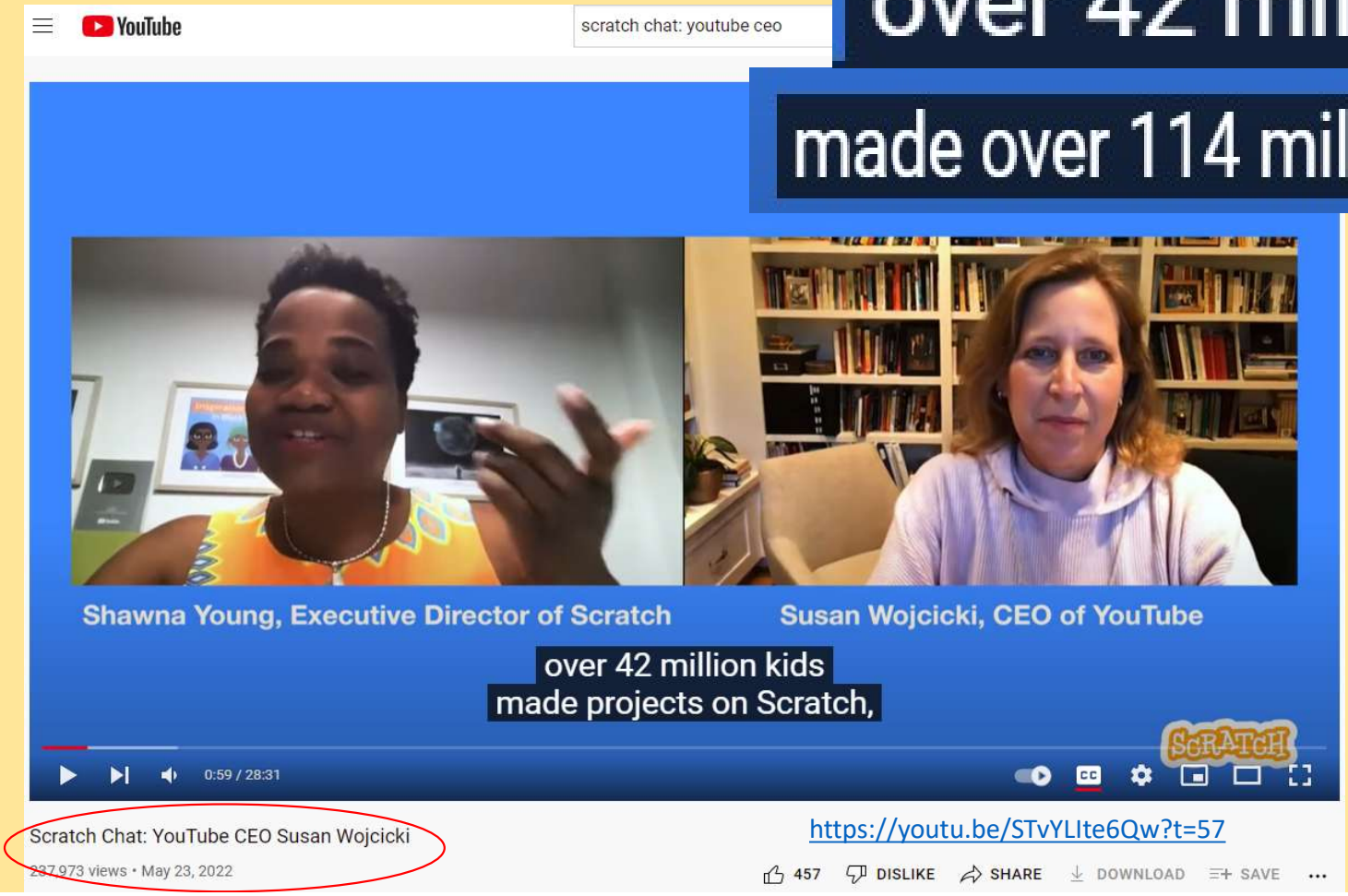
https://youtu.be/jXUZaf5D12A [1m:36s]

How many kids (Scratchers/Creators <- You) made projects on Scratch?

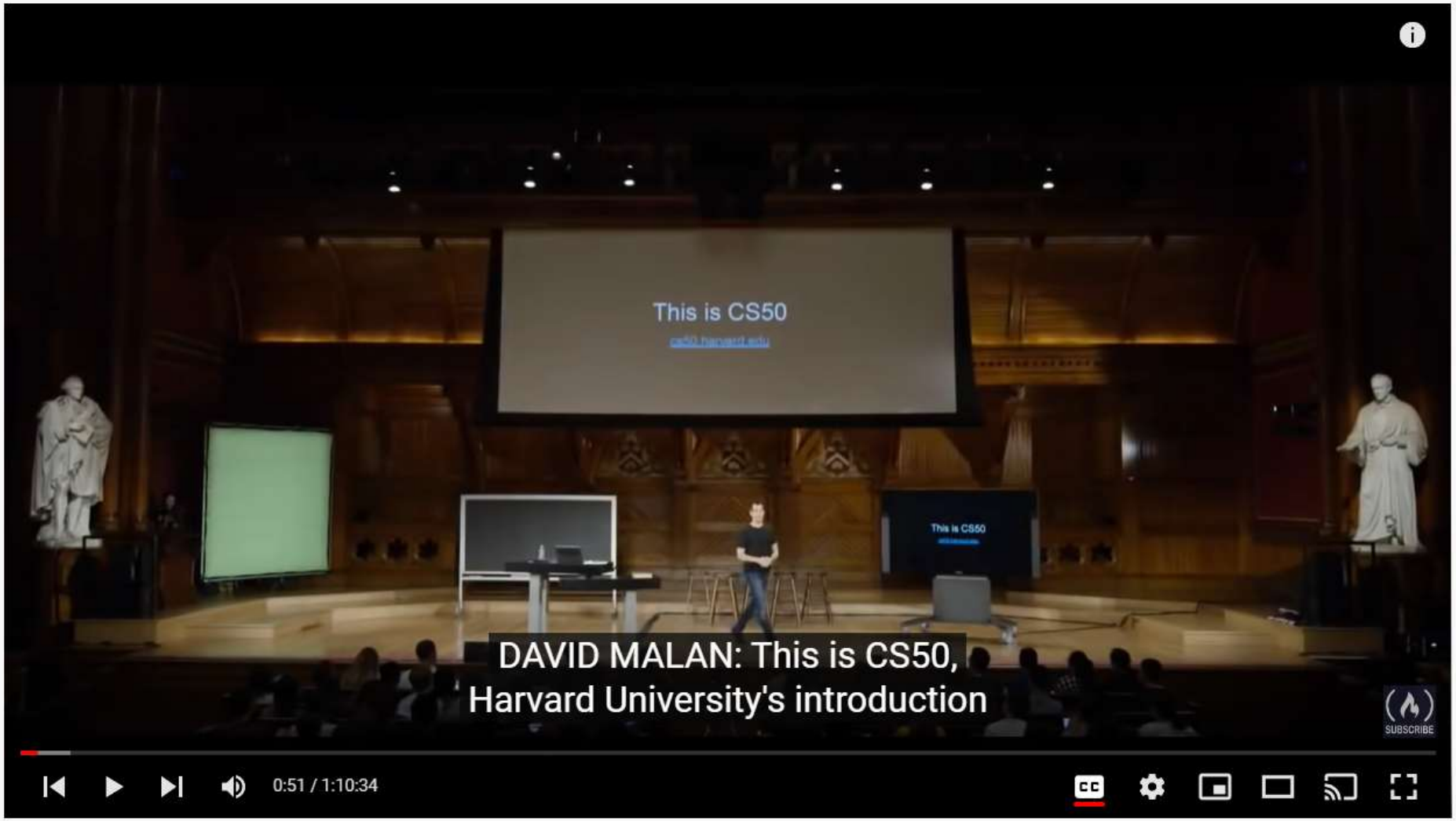
over 42 million kids

made over 114 million projects.

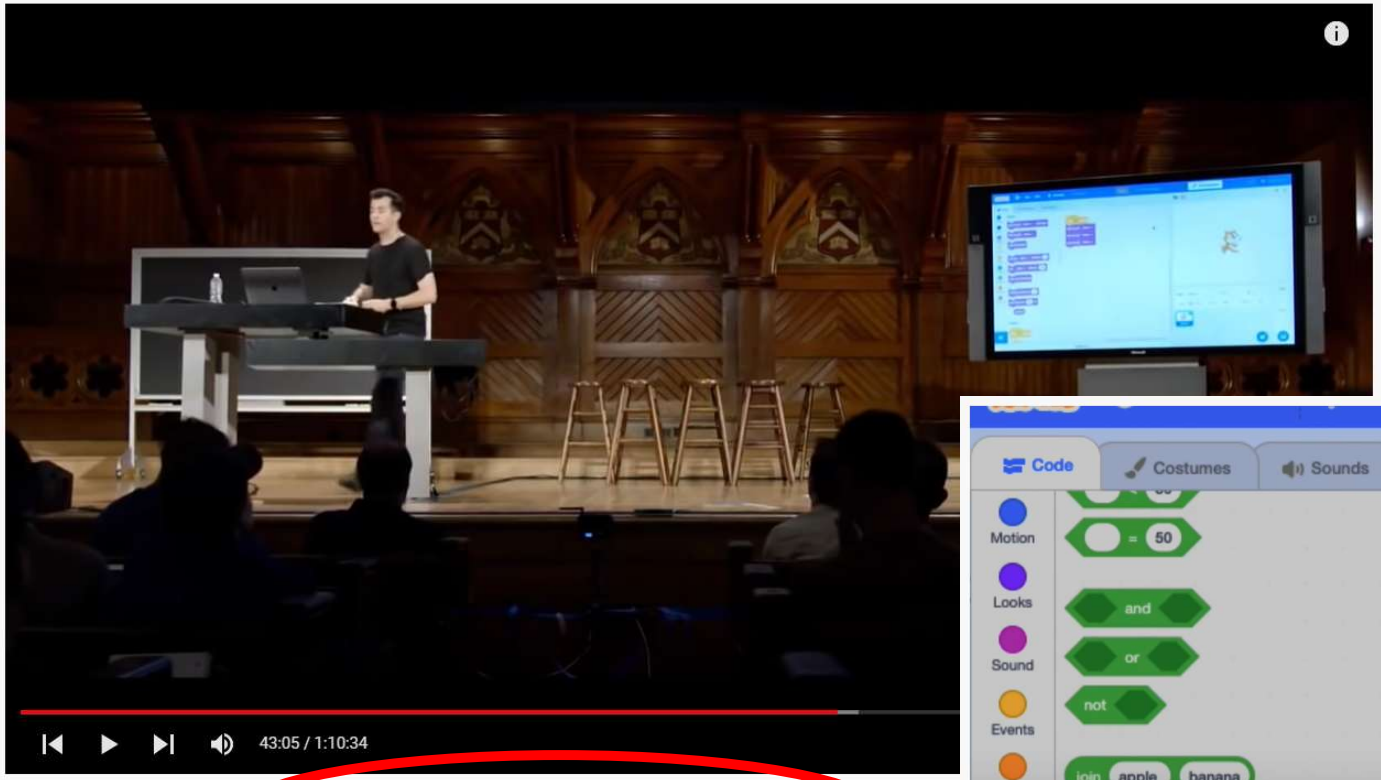


https://youtu.be/STvYLIte6Qw?t=57

It is never too late to get started!



Computational Thinking & Scratch - Intro to Computer Science - Harvard's CS50 (2018)

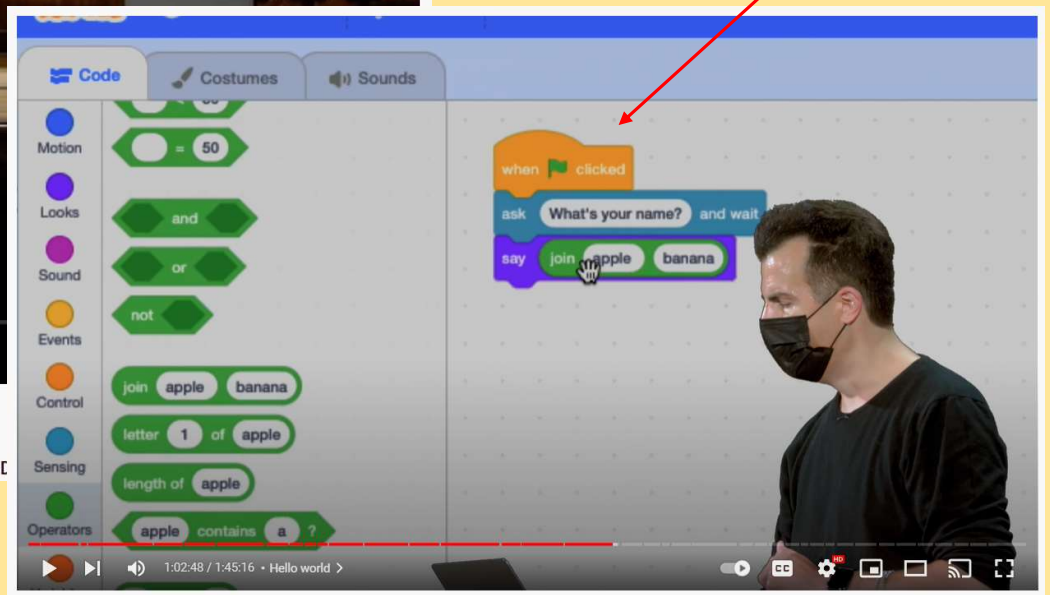


Computational Thinking & Scratch - Intro to Computer Science - Harvard's CS50 (2018)

795,063 views • Jan 22, 2019 11K DISLIKE DOWNLOAD

https://youtu.be/F0WoVEr0-44?t=2585

Harvard University – Scratch 3
CS 50 – by Dr. David Malan
Lecture 00



CS50 2021 in HDR - Lecture 0 - Scratch

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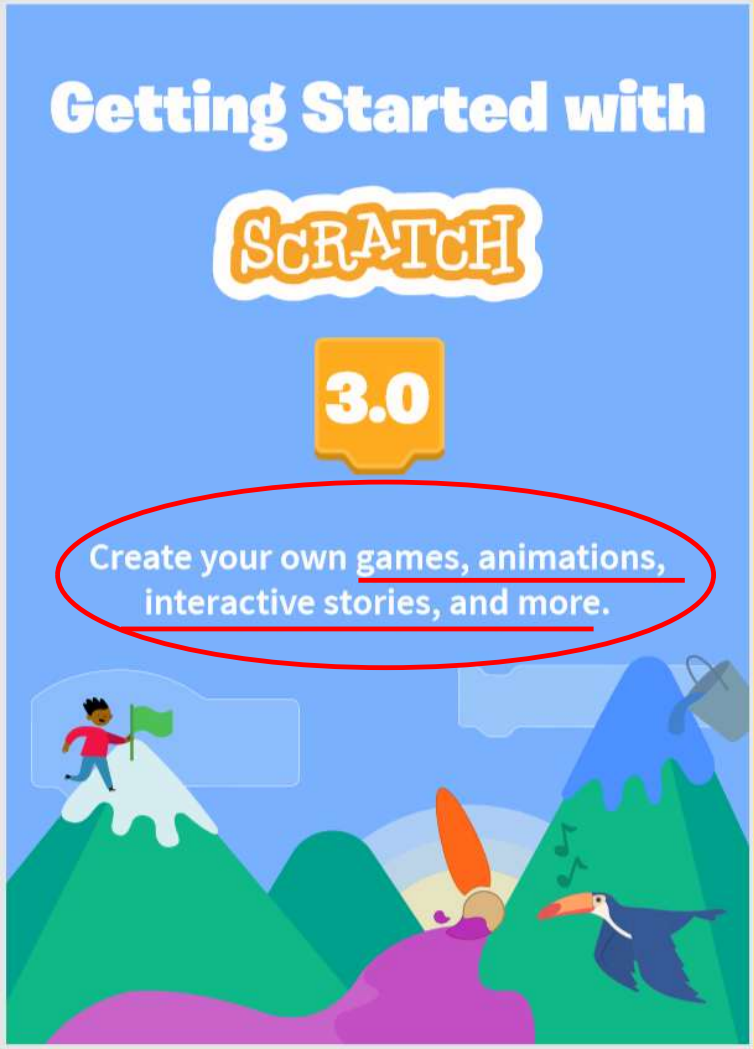
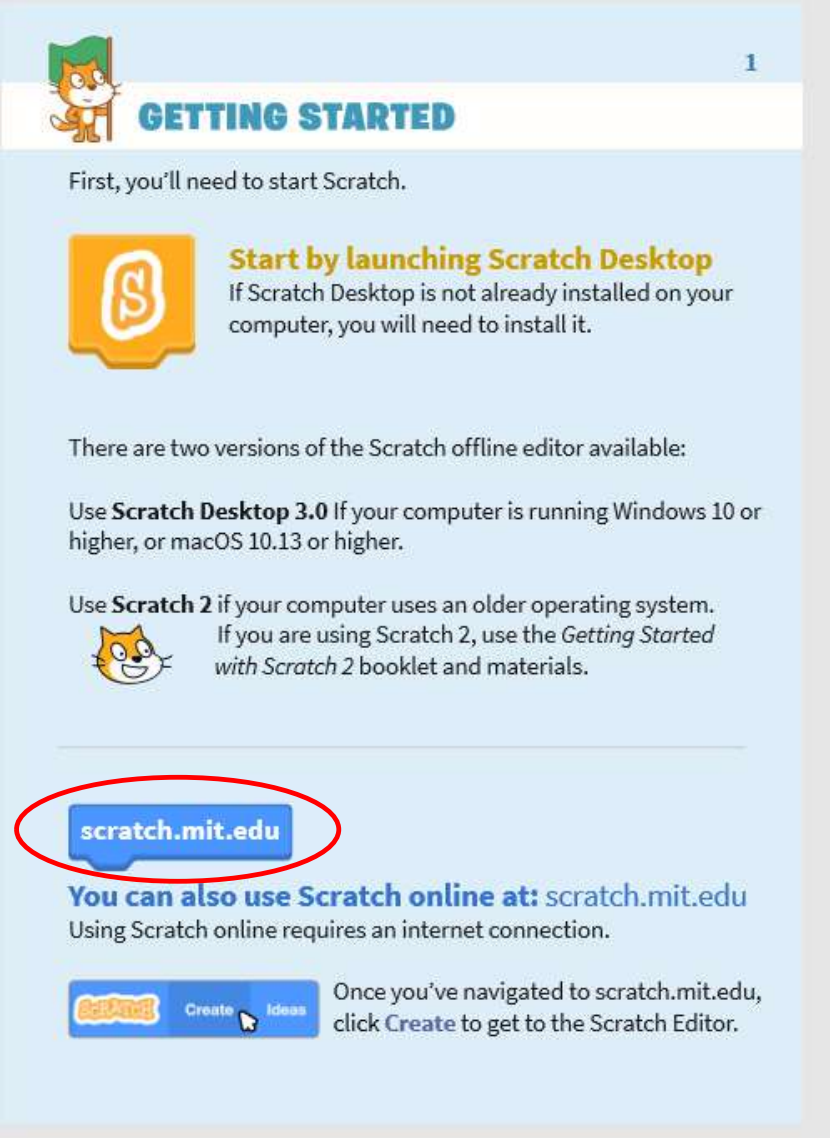
CS50





1

GETTING STARTED

First, you'll need to start Scratch.

Start by launching Scratch Desktop
If Scratch Desktop is not already installed on your computer, you will need to install it.

There are two versions of the Scratch offline editor available:

Use **Scratch Desktop 3.0** If your computer is running Windows 10 or higher, or macOS 10.13 or higher.

Use **Scratch 2** if your computer uses an older operating system. If you are using Scratch 2, use the *Getting Started with Scratch 2* booklet and materials.

scratch.mit.edu

You can also use Scratch online at: scratch.mit.edu
Using Scratch online requires an internet connection.

Create Ideas

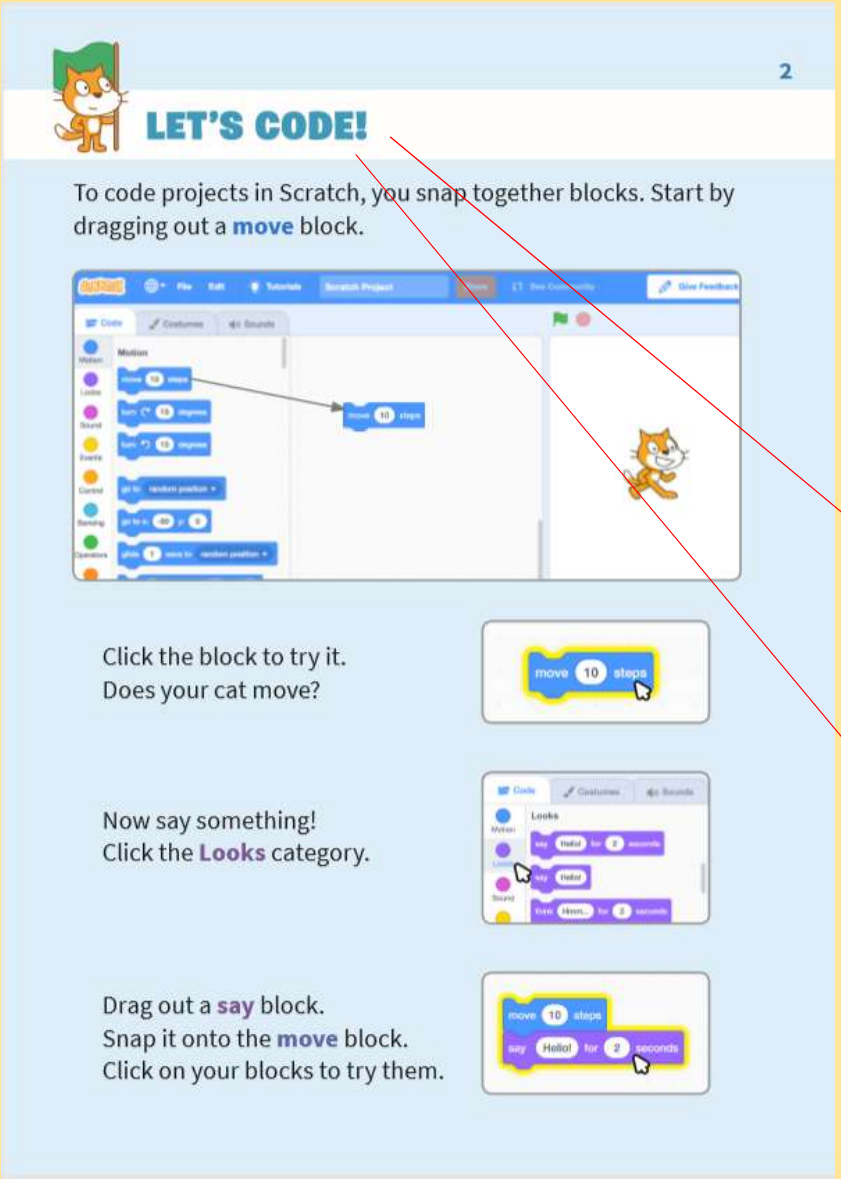
Once you've navigated to scratch.mit.edu, click **Create** to get to the Scratch Editor.

file:///H:/temp/IoT/Getting-Started-With-Scratch-3.0.pdf

User-Name and Password

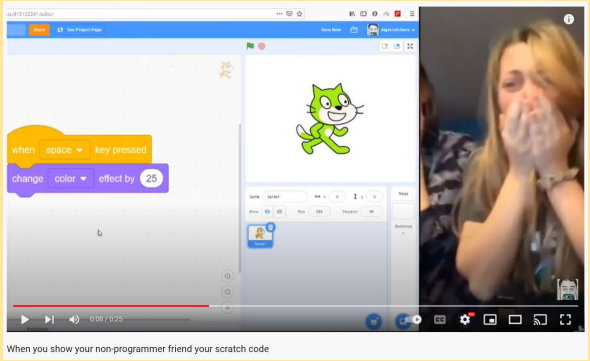
- **For Durant campus,**
  - **Login-name: seguest**
  - **Password: camp2022!**

- **For Idabel campus, please check with your co-host(s)**
- **For Marietta campus, please check with your co-host(s)**

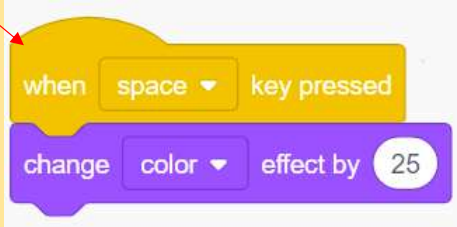


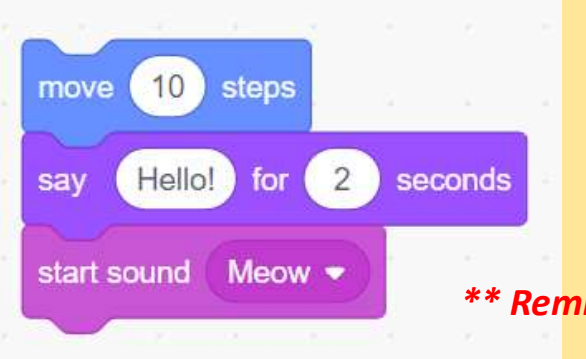
The Joy of coding …

When you show your non-programmer friend your scratch code



https://youtu.be/d0htdIq8J90 [0:25, Coding with Chris]





***** Reminder: Turn the speaker on!***

THE SCRATCH EDITOR

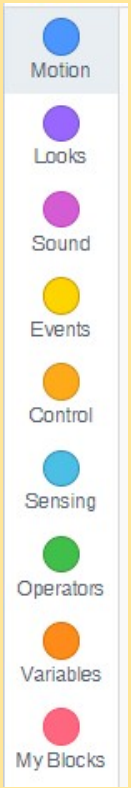
The Scratch Editor is where you create projects in Scratch. Here are its main parts:

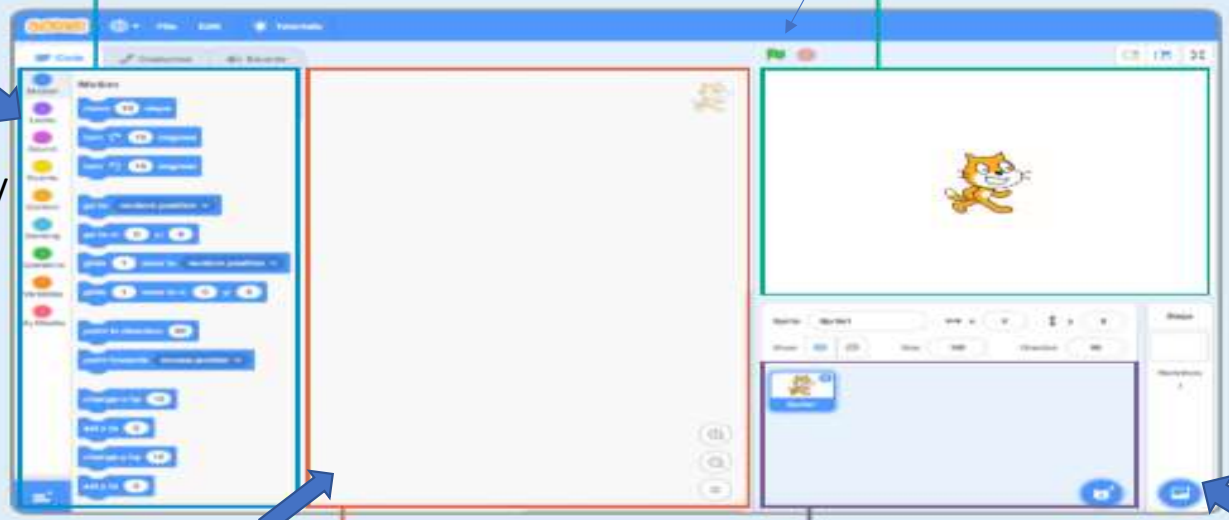
Saving in Scratch Desktop
Click the File menu and select **Save to your computer.** To open a project, click **Load from your computer.**

Blocks Palette
Blocks for coding your projects

The Stage
Where your creations come to life

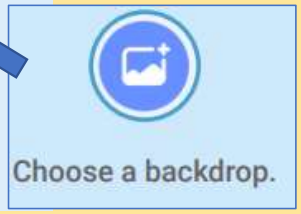
Menu/ Category/ Toolbox





Coding Area /script, playscript
Drag in blocks and snap them together to code your sprites

Sprite List
Click the thumbnail of a sprite to select it



WHAT IS A SPRITE?

Avatar, Actor, or ...

In Scratch, any character or object is called a sprite. Every new project in Scratch starts with the Cat sprite.

Want to choose a different sprite?

Click the New Sprite icon.

Or, hover over the "New Sprite" icon to see more options.

Upload an image from your computer.

Draw your own sprite.

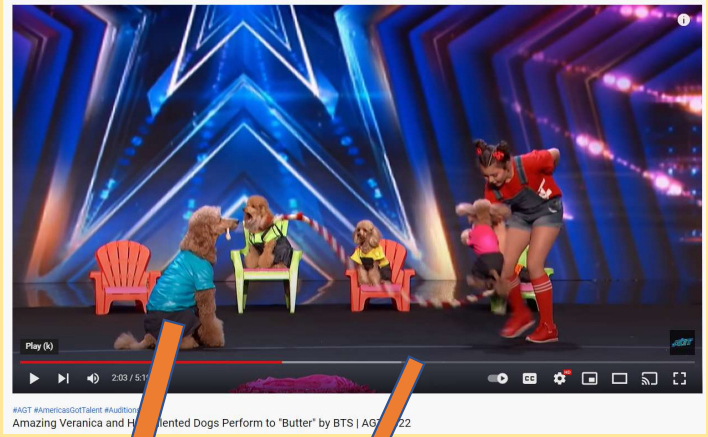
Click for a surprise sprite!

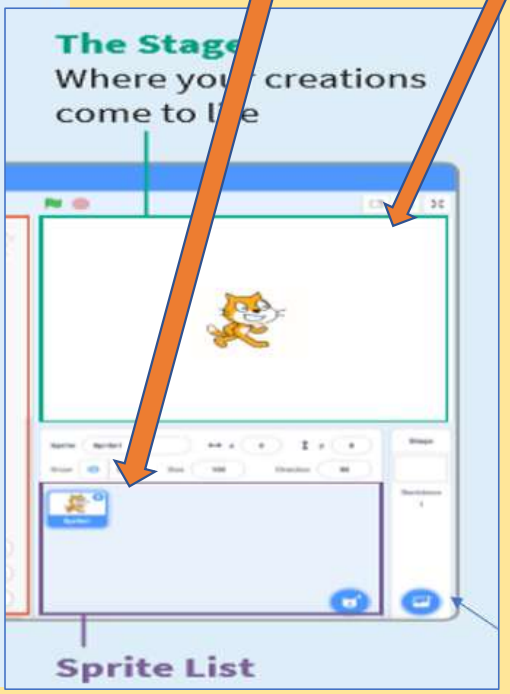
Choose a sprite from the library.

Want to **delete a sprite** from your project?

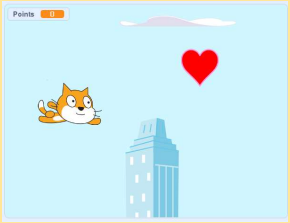
First, select the sprite by clicking on its thumbnail in the Sprite List.

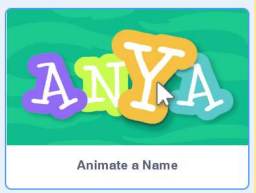
Then, click here to delete the sprite.





Objects:
heart, building, cloud, letters

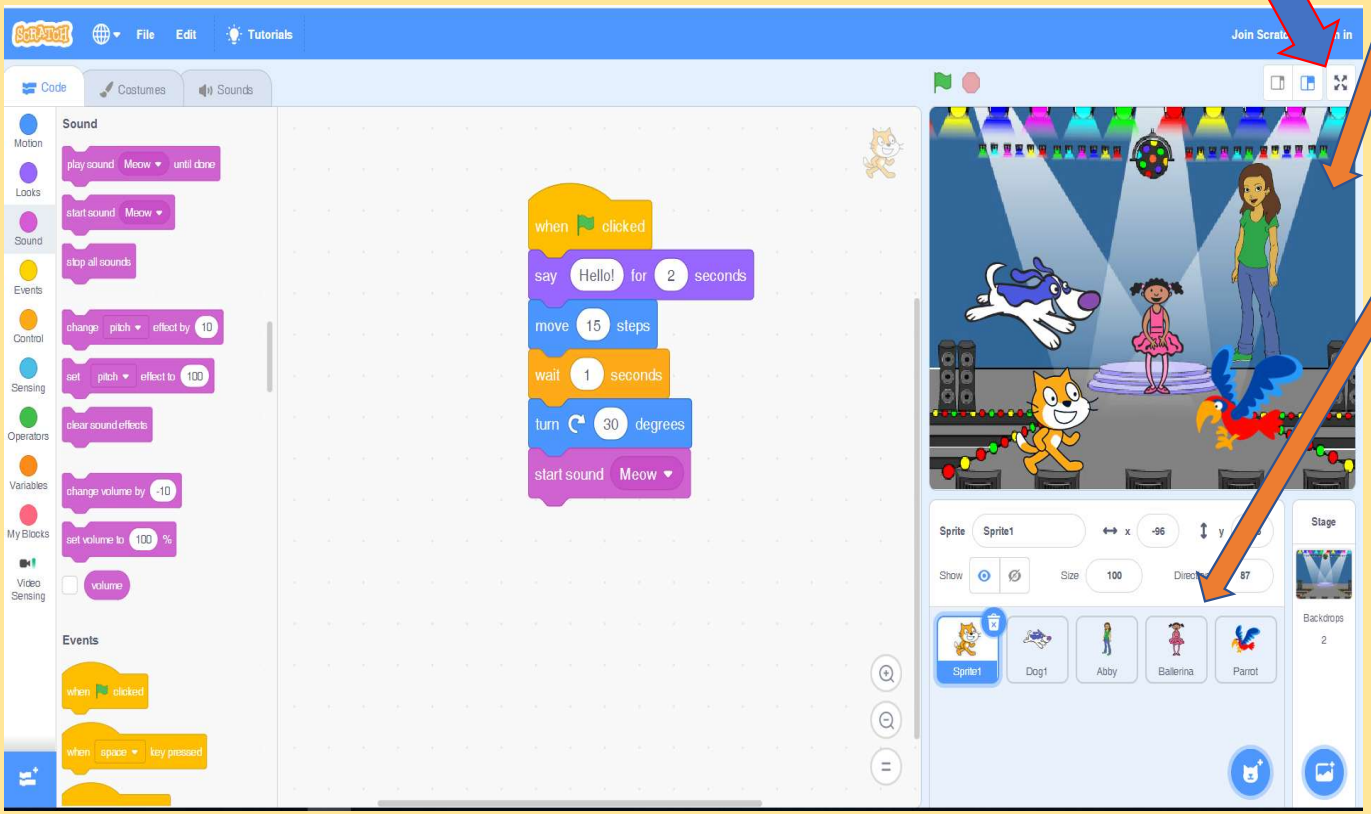


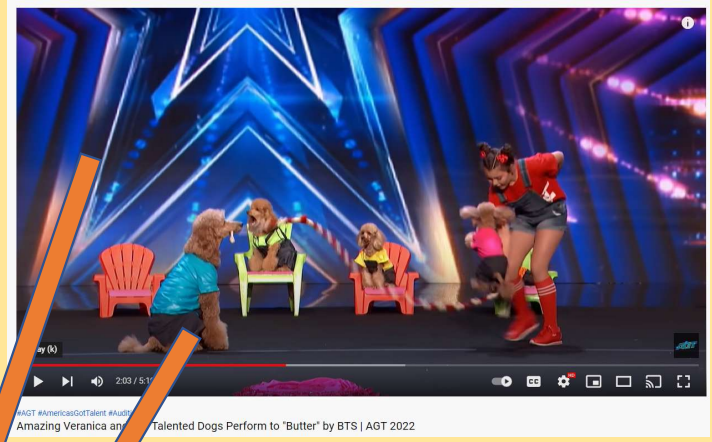


Hands-on:

Learn by doing it - Can we try it?

- Add some Sprites
- Add a Backdrop/Background





https://youtu.be/OzcnneNlOtI?t=123

WHERE IS YOUR SPRITE?

Every sprite has an **x** and **y** position on the Stage.

x is the position of the sprite from left-to-right.

y is the position from top-to-bottom.

At the very center of the stage, **x** is 0 and **y** is 0.

When you move your sprite, you can see its **x** and **y** position change.

GET CREATIVE!

Encourage students to use their imagination as you create projects. There are many different ways they can make their Scratch projects unique.

You can choose or draw your own characters.

Choose a sound or record your own.

Try changing numbers or adding blocks to your code to see what happens.

Experiment and customize your project however you want!

- **Any Questions?**

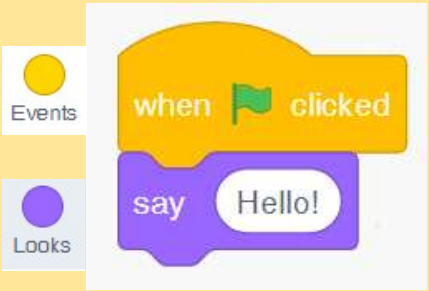
Scratch – the basic concepts of coding

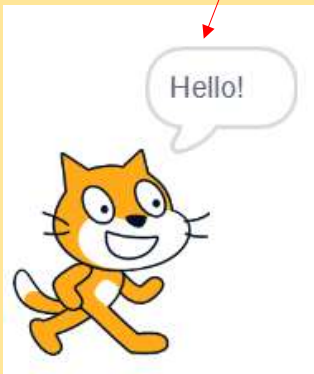
- Basic computer inputs and outputs
- Declaring variables
- Arithmetic Operation (+, -, *, /)
- Control structures
  - Sequence
  - Repetition (Looping/iteration)
  - Selection (Condition)
- Events (key pressed, mouse clicked, sprites overlapping, …)
- Other topics (cloning, broadcast message, ..)

- Creating **hands-on game projects**.

Basic Input (keyboard, mouse, …) / Basic Output (text, sound, color, …)

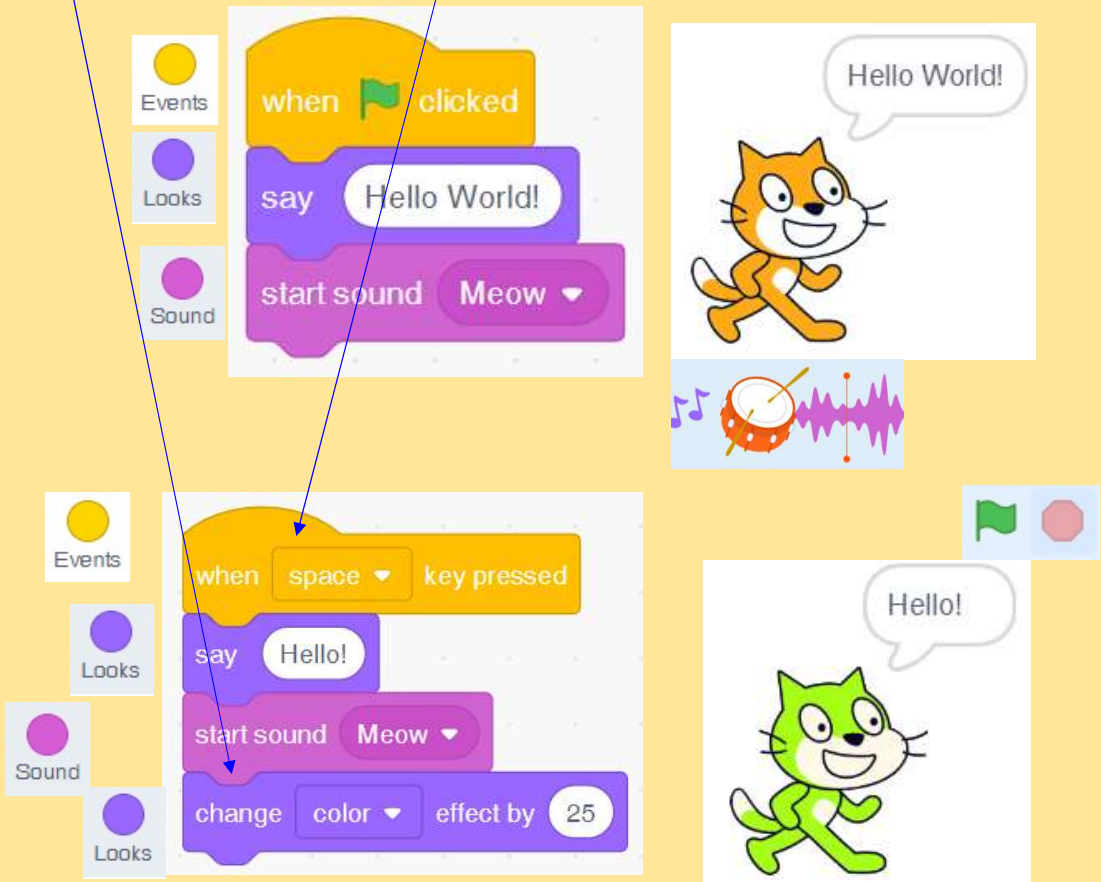
"Hello World" – 50 programming languages





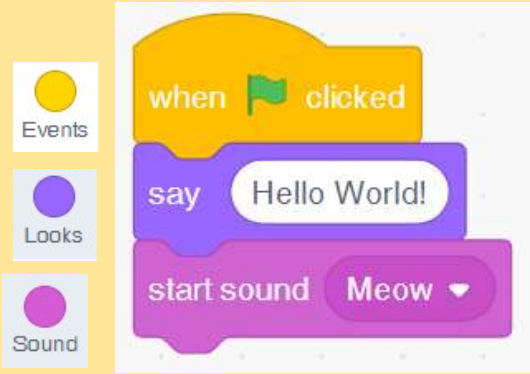
Basic Input (keyboard, mouse, …) / Basic Output (text, sound, color, …), + Event (keyPressed)

"Hello World" – 50 programming languages

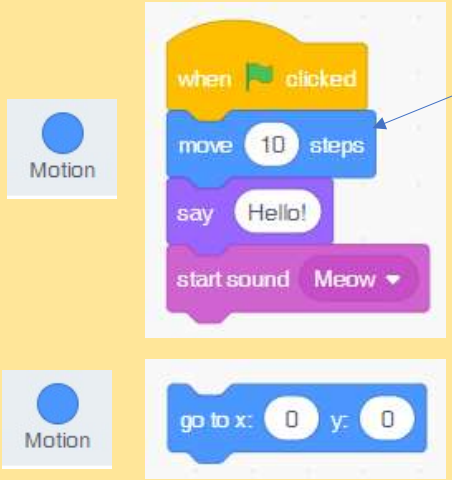


Basic Input (keyboard …) / Output (text, sound, movement…)

Sequencing … code-blocks



Move the Cat forward?



Move the Cat backward?
Too fast?

Move the Cat backward?
Wait until sound done
Move backward

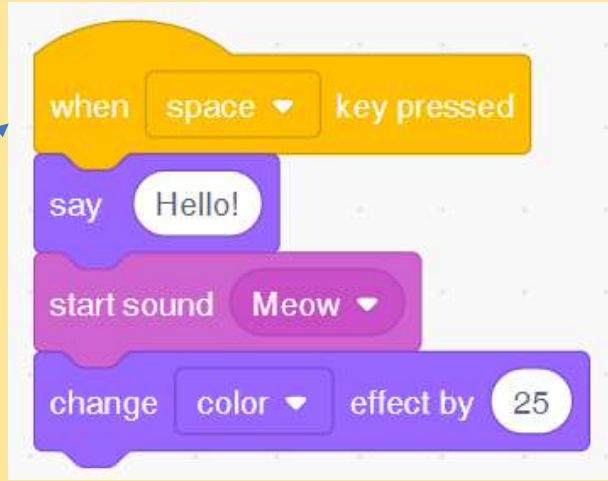
Algorithm:
a process or set of rules (instructions) to be followed in calculations or other problem-solving operations, especially by a computer. - Oxford Languages

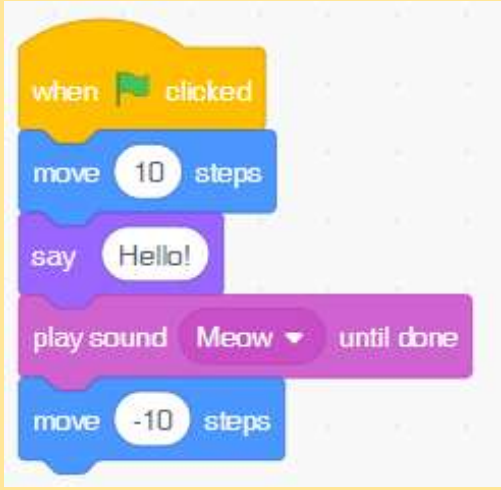
Or a Recipe: the steps/instructions to cook a dish!

Algorithm:

a. **Move 10 steps forward**
b. **Say "Hello!"**
c. **Play sound "Meow"**
d. **Move 10 steps backward.**

- **Basic Input**
  - **keyboard, mouse, …**
- **Basic Output**
  - **text, sound, color, movement, …,**
- **Sequence (code-block executed in sequence, top-to-bottom)**
- **Event (keyPressed: input)**





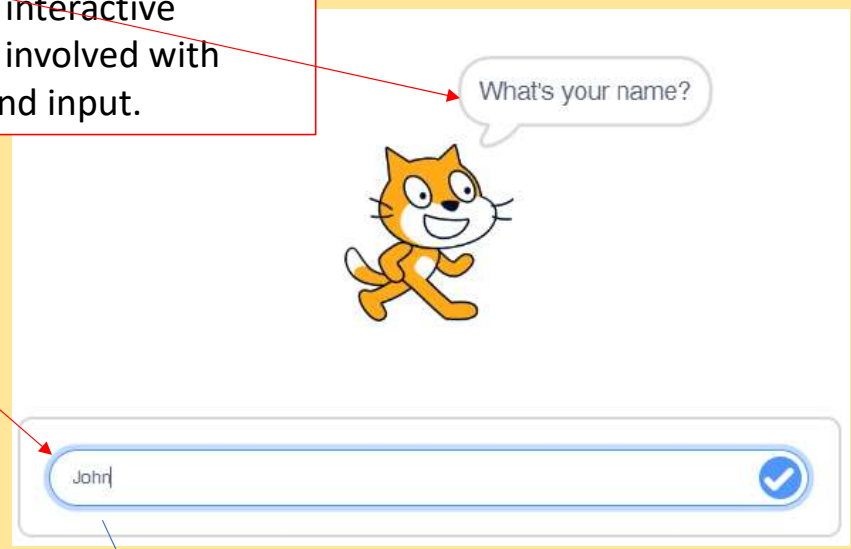
Scratch – the basic concepts of coding

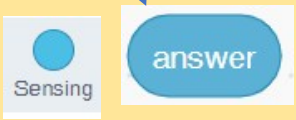
- Basic computer inputs and outputs
- Declaring variables
- Arithmetic Operation (+, -, *, /)
- Control structures
  - Sequence
  - Repetition (Looping)
  - Selection (Condition)
- Events (key pressed, mouse clicked, sprites overlapping, …)
- Other topics (cloning, broadcast message, ..)

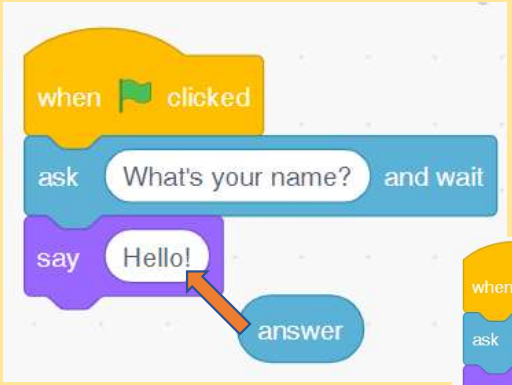
- Creating **hands-on game projects**.

Basic Input / Output, Variable - Sequencing, T-2-B)

A simple interactive program involved with output and input.



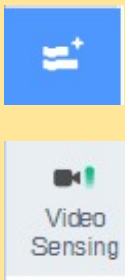


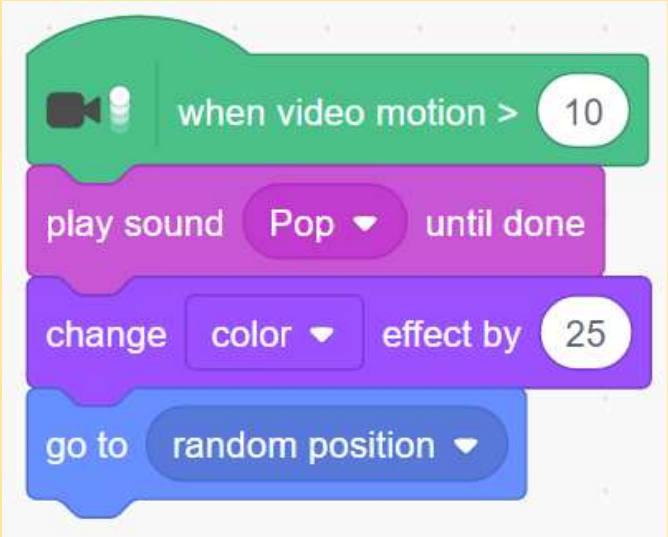


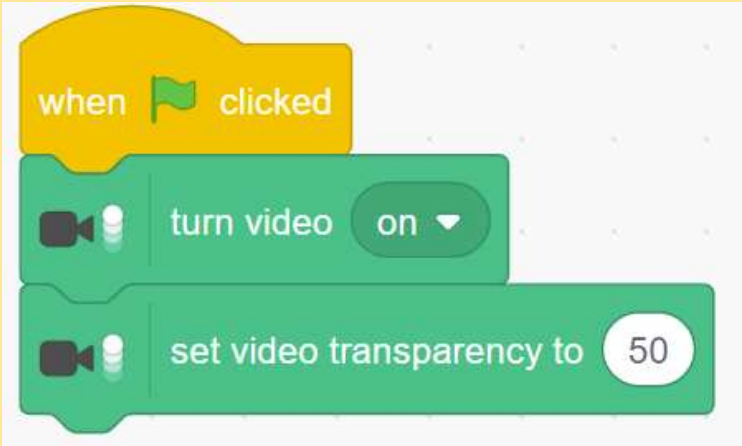
- Be aware that the "Hello" won't get displayed until the check-mark is clicked or enter-key is pressed.
- Why?
- The "wait" – until check-mark is clicked
- Where the input-value (e.g. John) is stored?

  - answer (a build-in variable)

Game: Pop the Balloon (needs a webcam or camera)

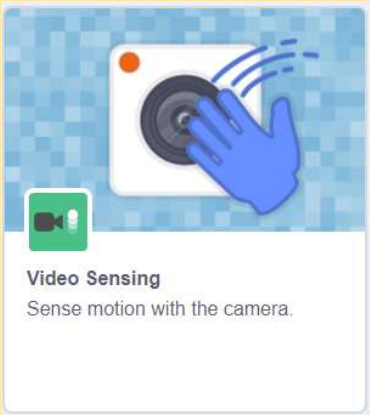






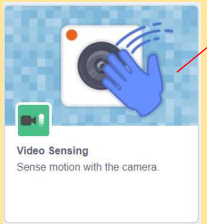
https://youtu.be/R1ezjDKiJVg?t=16

***** Reminder: Have the Webcam/Camera ready!***



Game: Pop the Balloon

(Application on variable: we want to keep the **Score**. But How?)



when video motion > 10
play sound Pop until done
change color effect by 25
go to random position

Variables

Variables
Make a Variable
my variable

New Variable
New variable name:
Score
For all sprites
For this sprite only
Cancel
OK

Score 0

when video motion > 10
play sound Meow until done
change color effect by 25
go to random position
change Score by 1

when clicked
turn video on
set video transparency to 50

Variables

when clicked
set Score to 0

Reset Score back to 0

Scratch – the basic concepts of coding

- Basic computer inputs (Camera) and outputs
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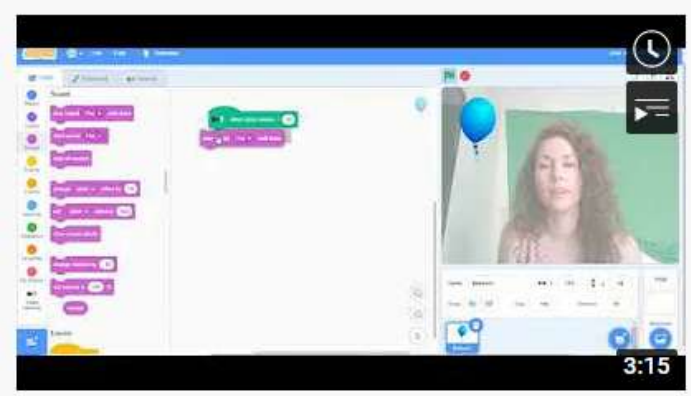
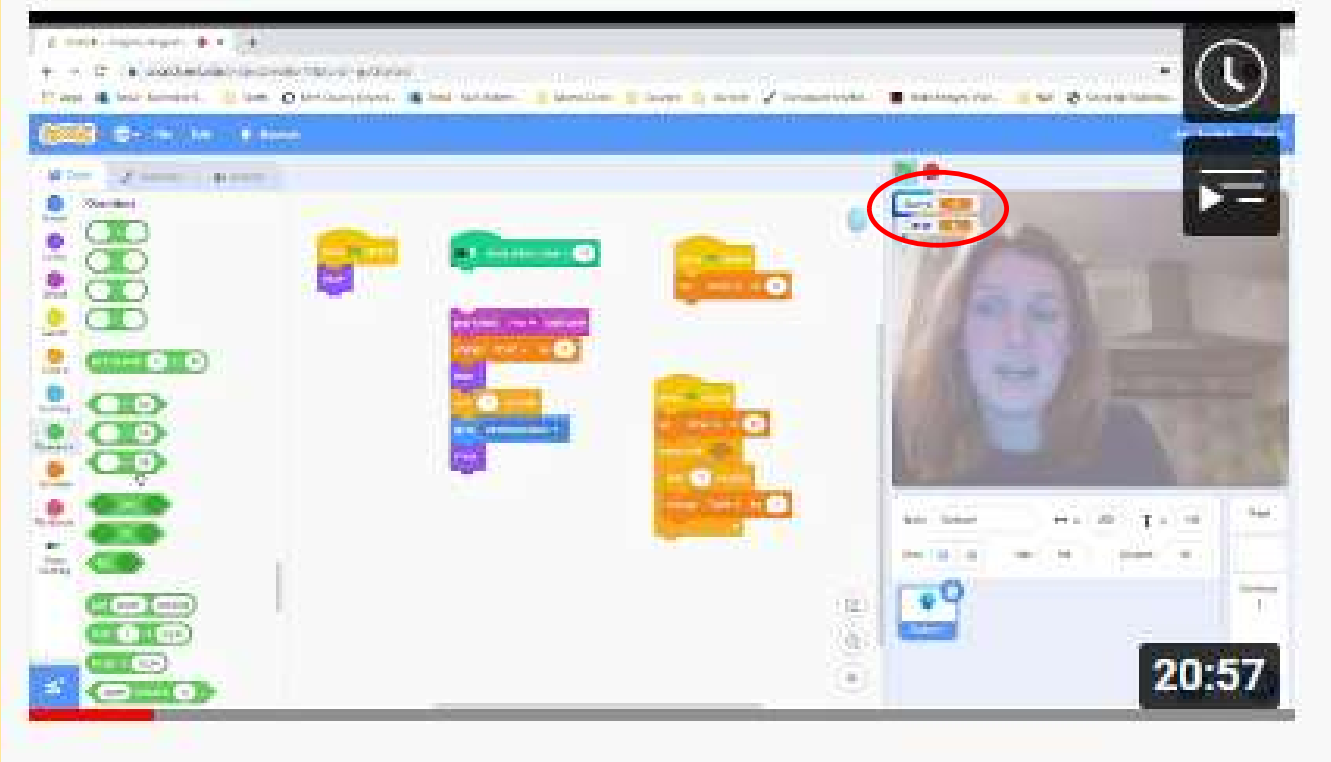
  - Repetition (Looping)
  - Selection (Condition)
- Events (key pressed, mouse clicked, sprites overlapping, …)
- Other topics (cloning, broadcast message, ..)

- Creating **hands-on game projects**.

How to Pop Balloons using Video Sensing Extension on Scratch



https://youtu.be/23xstN6hgKI [3:14]



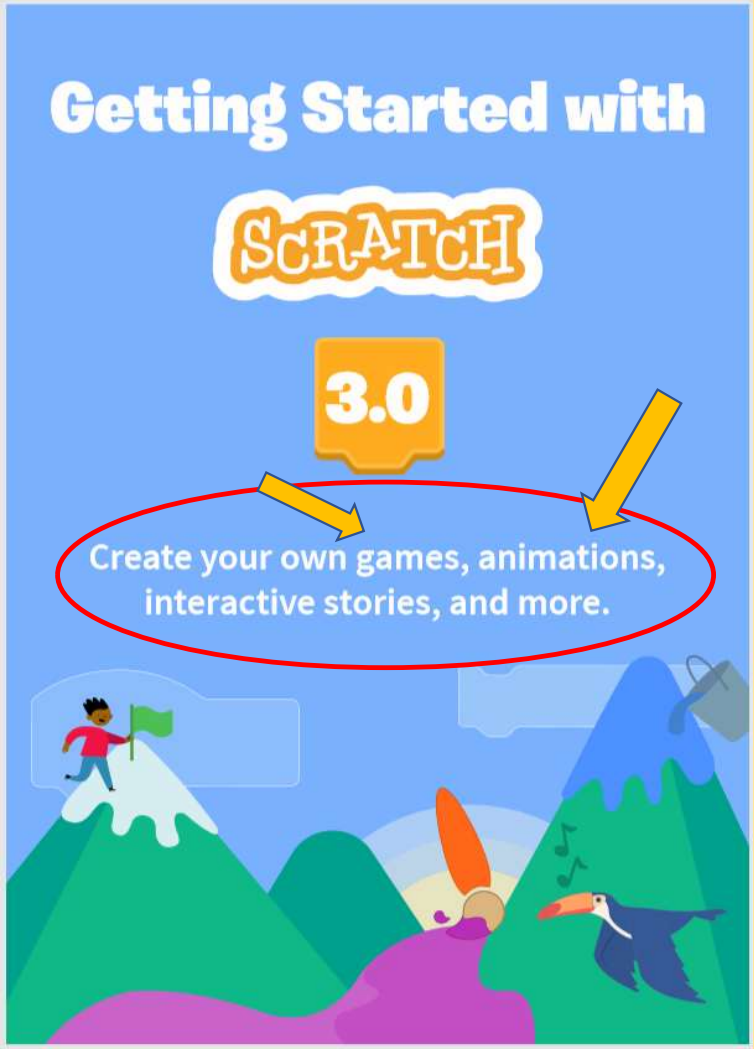
How to do video sensing in Scratch by Mrs Robertson

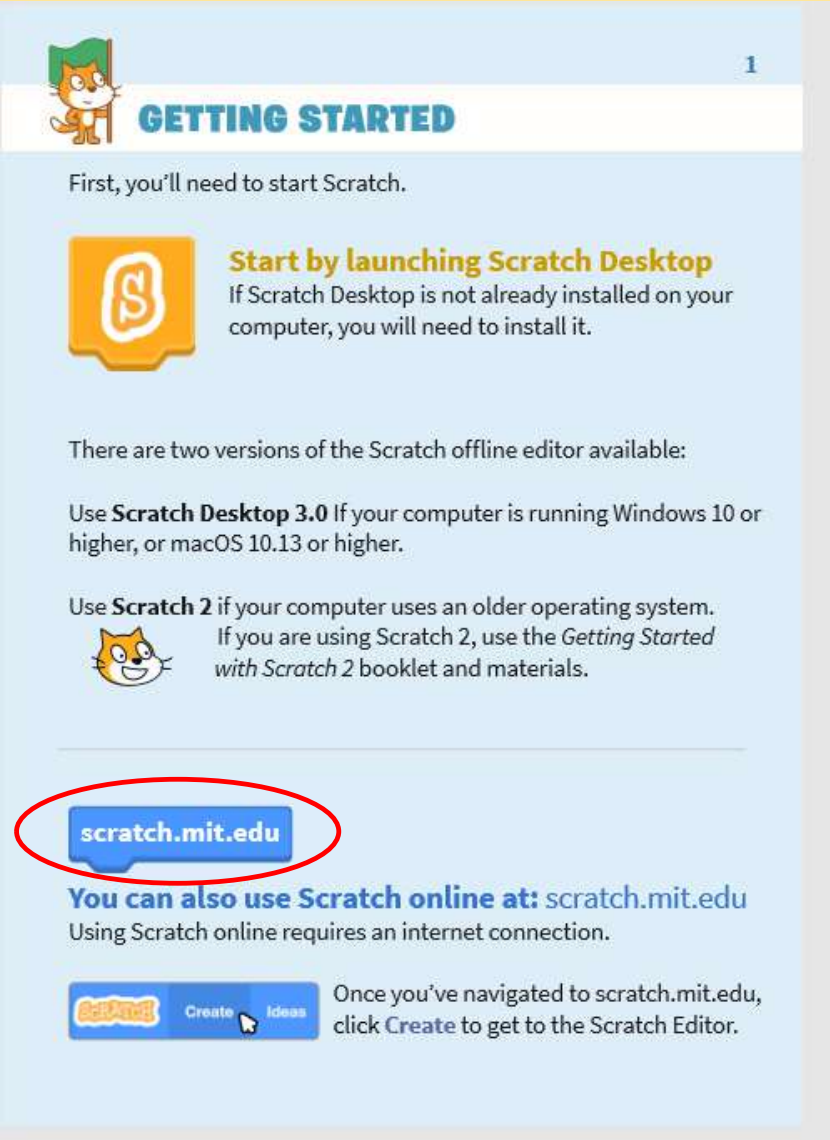
7.5K views • 1 year ago

Burstow Primary School

https://youtu.be/7JS1mkJQy2A?t=692 [20:56]

- Questions?
- Fun – Pop the balloon game!
- A Break?





1

GETTING STARTED

First, you'll need to start Scratch.

Start by launching Scratch Desktop
If Scratch Desktop is not already installed on your computer, you will need to install it.

There are two versions of the Scratch offline editor available:

Use **Scratch Desktop 3.0** If your computer is running Windows 10 or higher, or macOS 10.13 or higher.

Use **Scratch 2** if your computer uses an older operating system. If you are using Scratch 2, use the *Getting Started with Scratch 2* booklet and materials.

scratch.mit.edu

You can also use Scratch online at: scratch.mit.edu
Using Scratch online requires an internet connection.

Once you've navigated to scratch.mit.edu, click **Create** to get to the Scratch Editor.

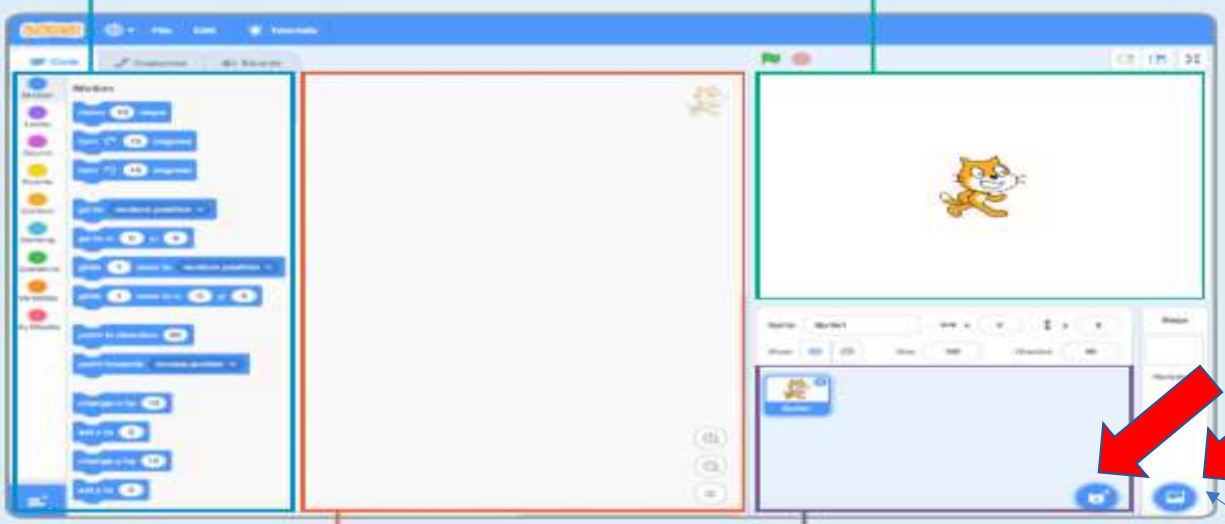
file:///H:/temp/IoT/Getting-Started-With-Scratch-3.0.pdf

THE SCRATCH EDITOR

The Scratch Editor is where you create projects in Scratch. Here are its main parts:

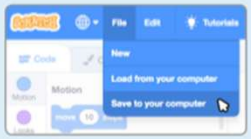
Blocks Palette
Blocks for coding your projects

The Stage
Where your creations come to life



Coding Area /script, playscript
Drag in blocks and snap them together to code your sprites

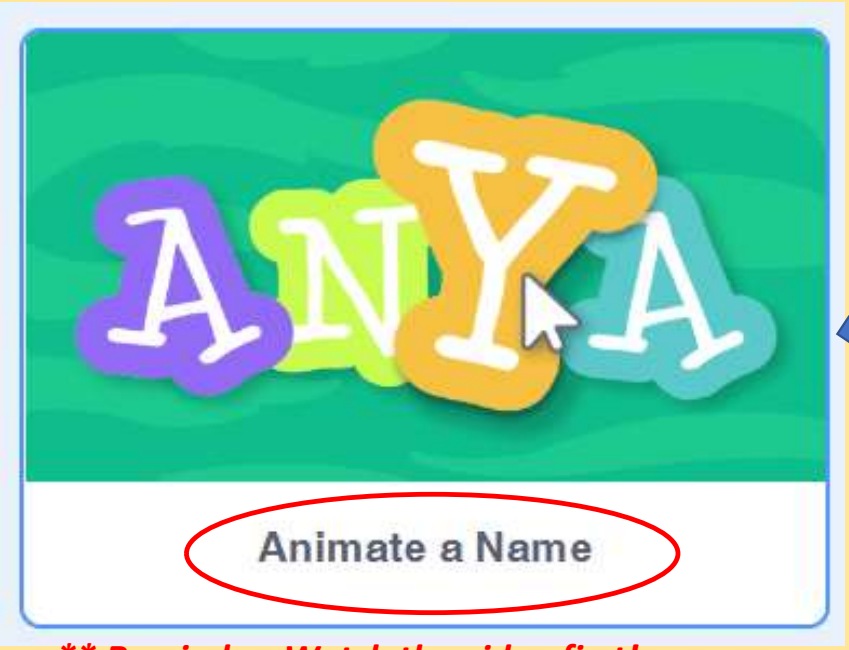
Sprite List
Click the thumbnail of a sprite to select it



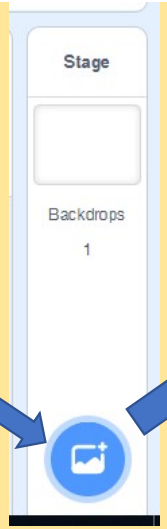
Saving in Scratch Desktop
Click the File menu and select **Save to your computer**. To open a project, click **Load from your computer**.

Let's add a new Backdrop (background) and more Sprites

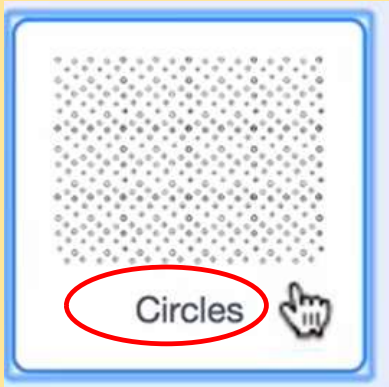
Choose a backdrop.

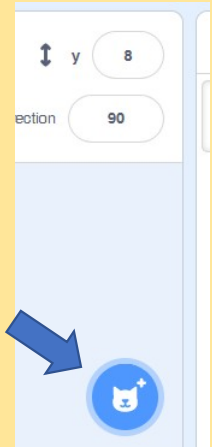


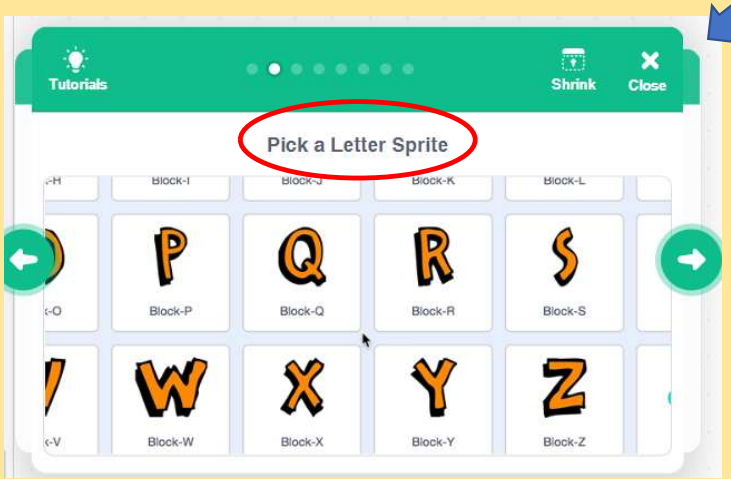
*** Reminder: Watch the video first!*

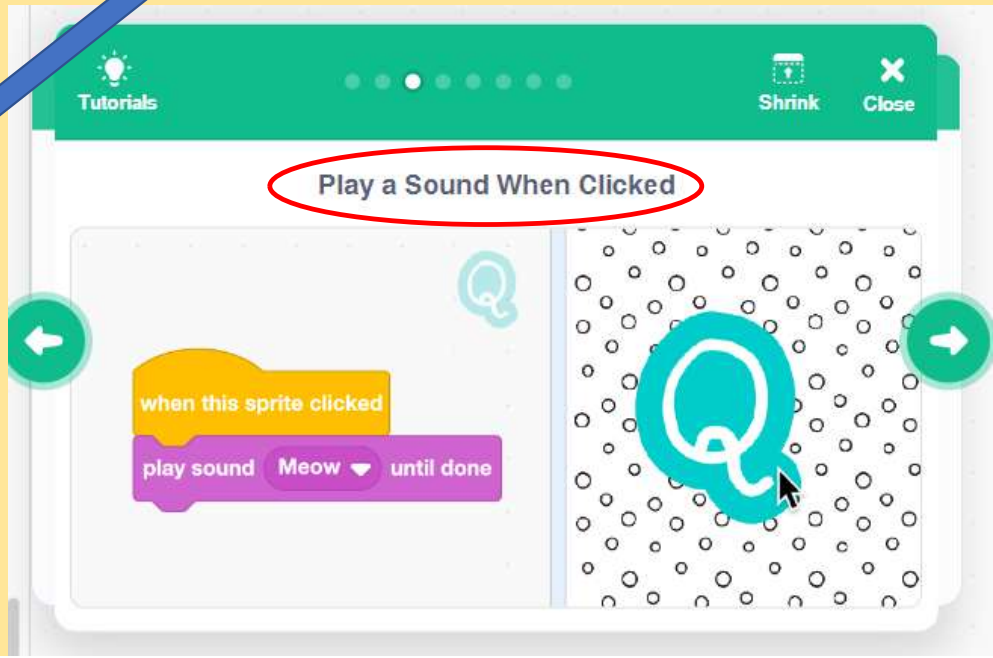


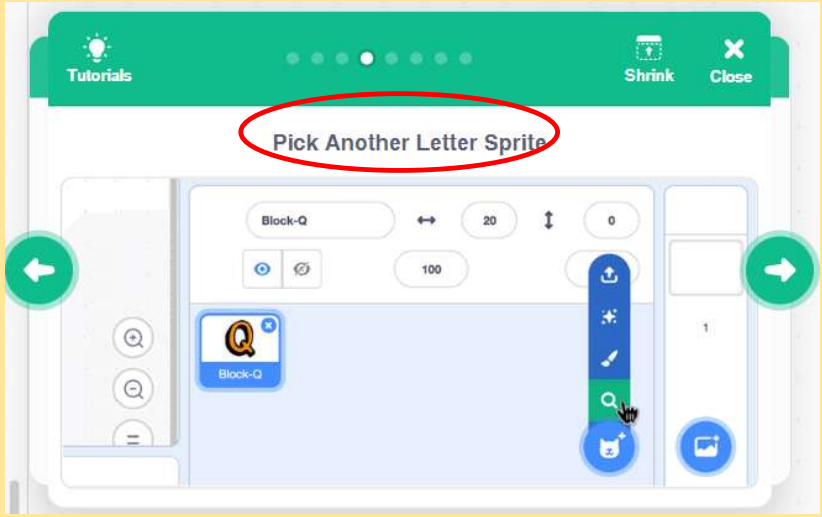
Backdrop or Background or Stage Scene

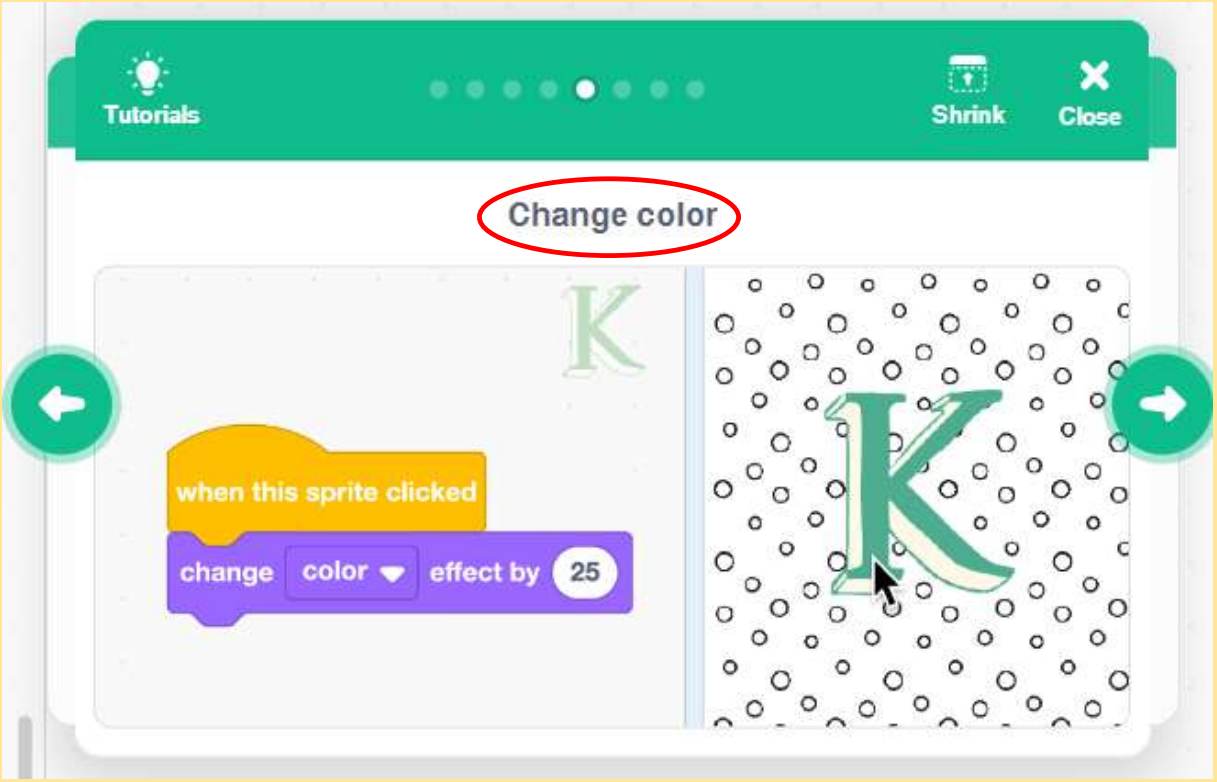


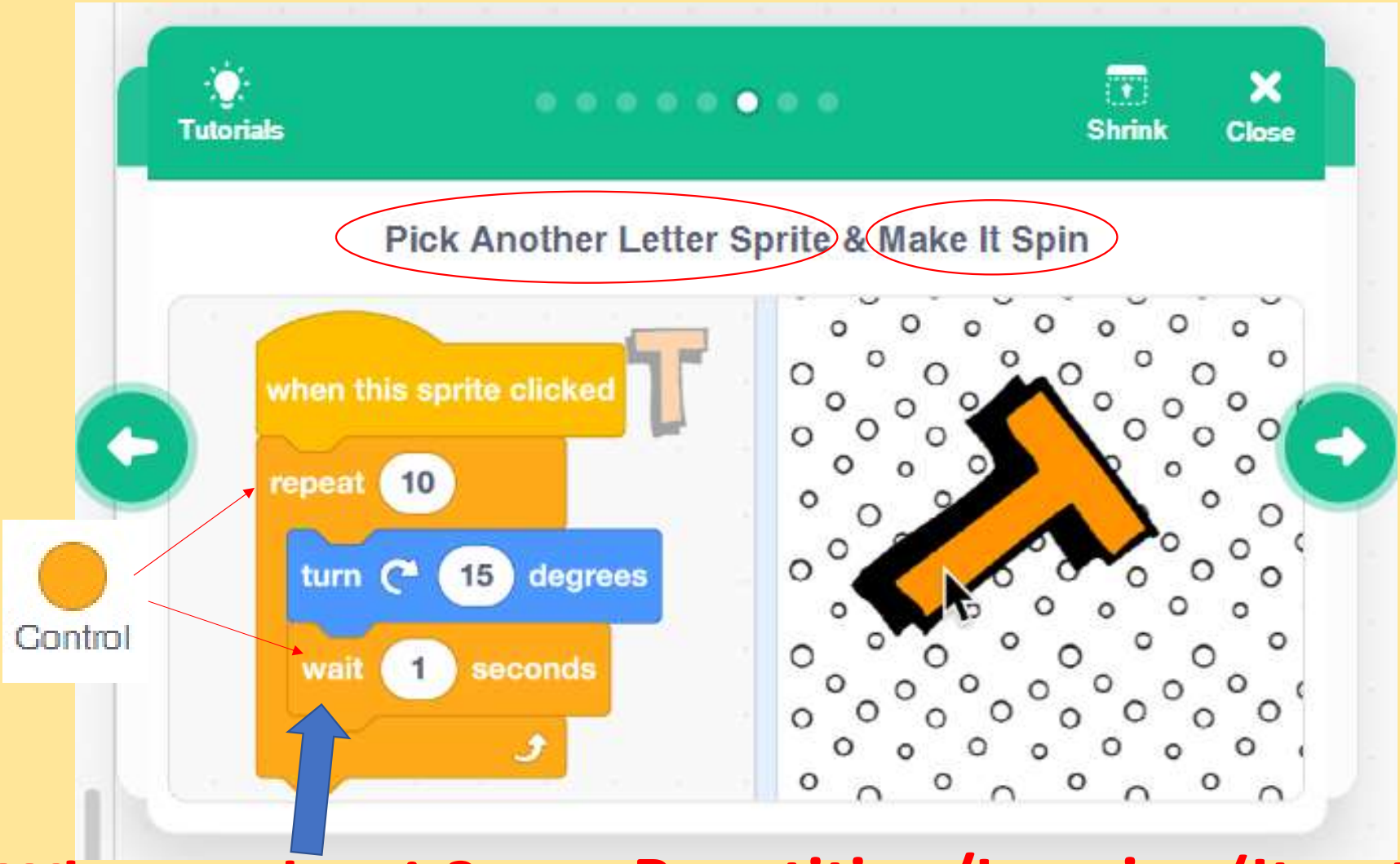




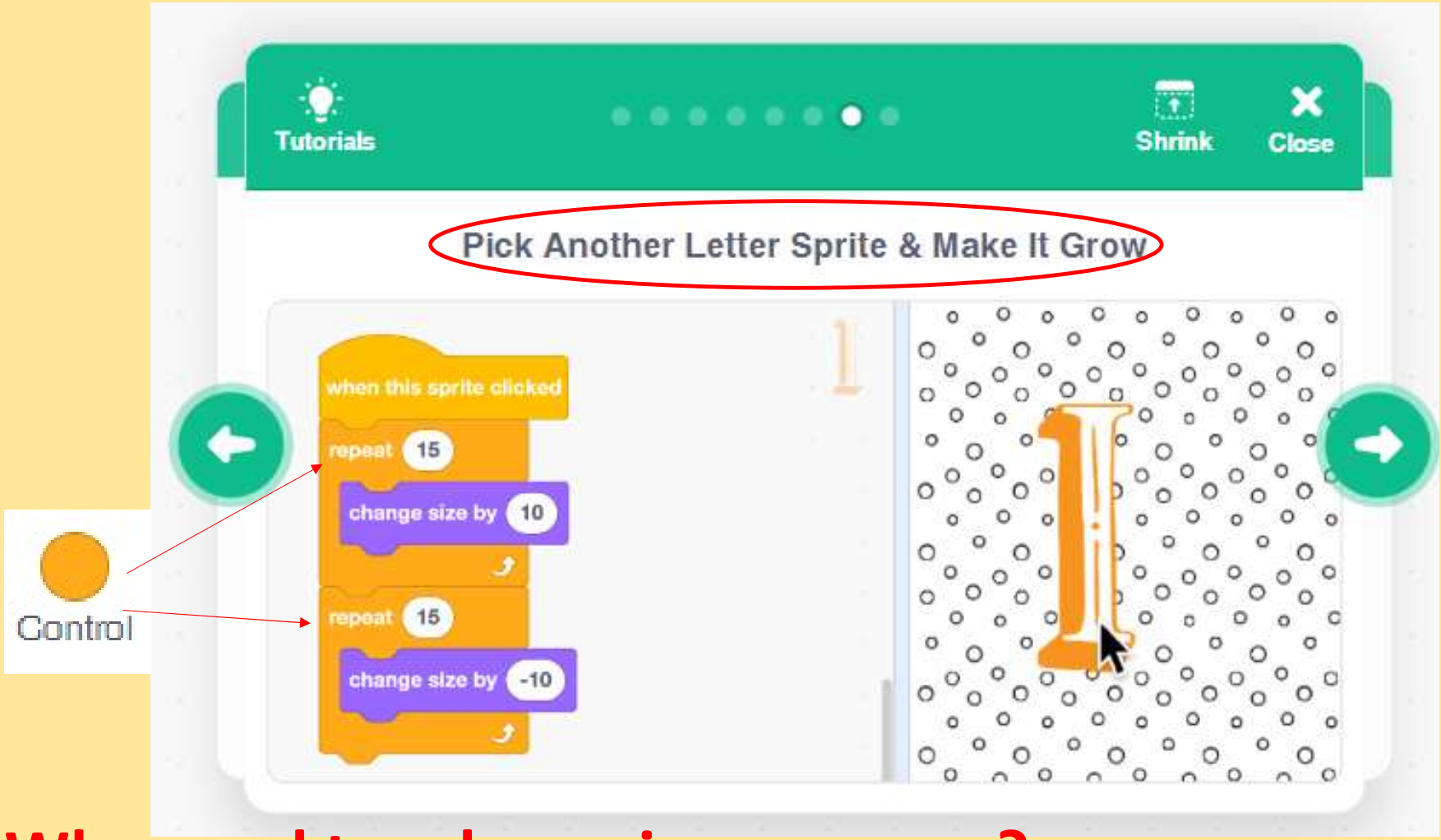



Tutorials
Shrink
Close
Pick Another Letter Sprite
Block-Q
20
0
100
Block-Q


Tutorials
Shrink
Close
Change color
when this sprite clicked
change color effect by 25



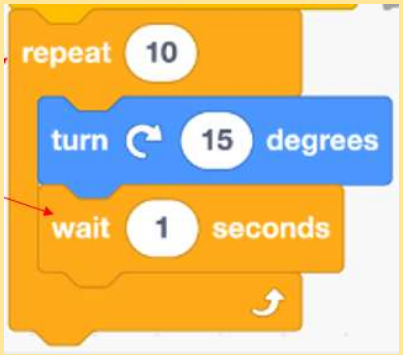
Why need wait? **Repetition/Looping (Iteration)**



Why need two loops in sequence?

What did we learn from this project?

- Animation
- Adding Backdrop (background, scene, stage, …)
- Adding New Sprites
- Repetition/Looping/Iteration
  - Rotating/Turing, Growing/Shrinking, …
- Waiting
- Event (when Sprite is clicked …)
- Two loops in sequence
- Know-How to use tutorial (I have the know-how, iPhone, but he)



Examples!

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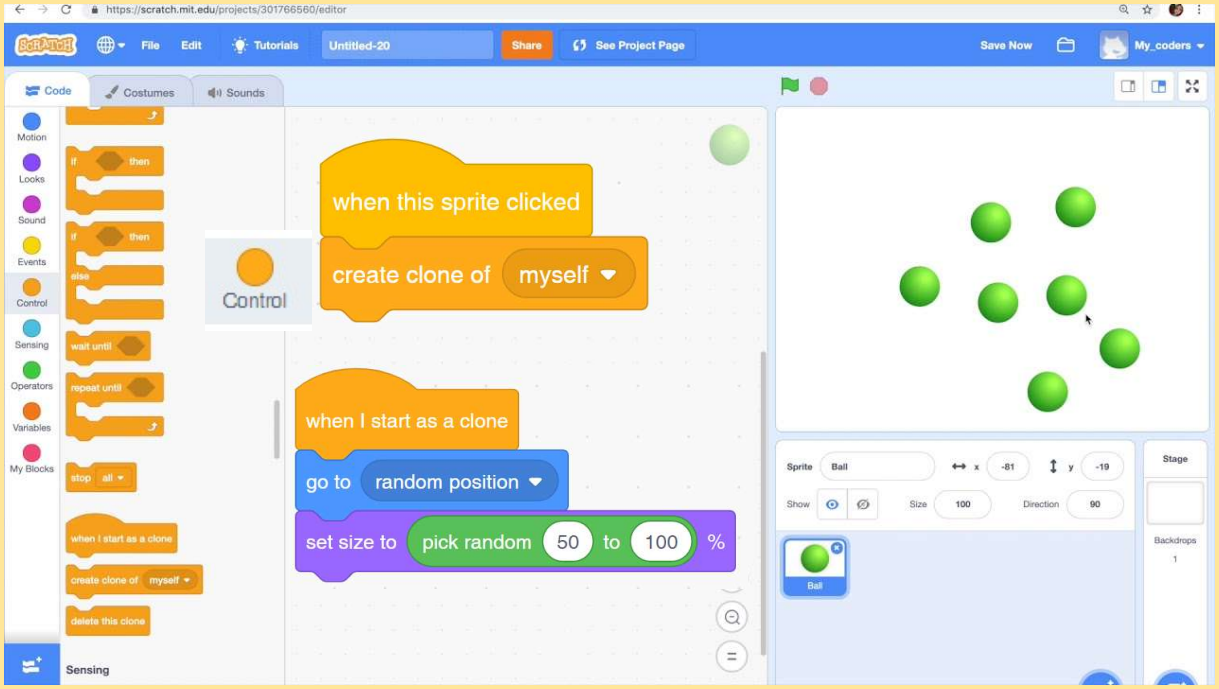
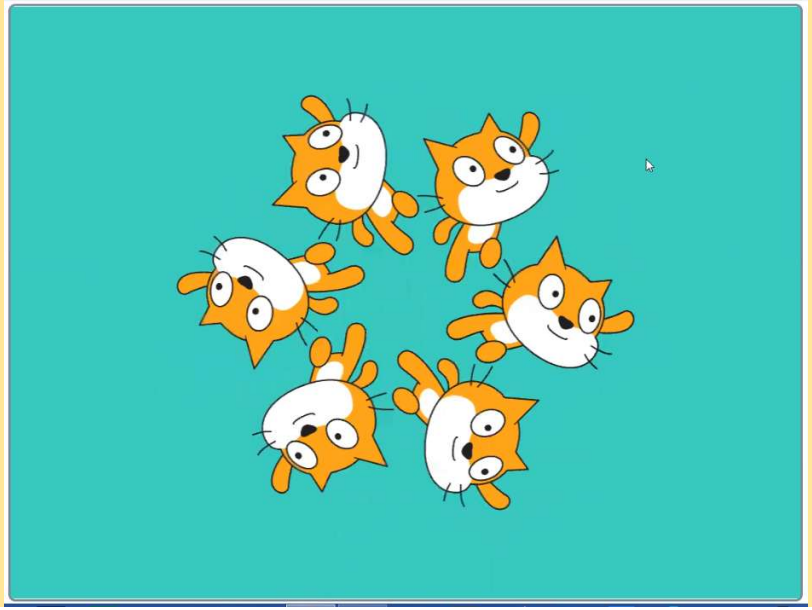
Create CLONES in Scratch



https://youtu.be/ihaCsZq3nEg [2:40]



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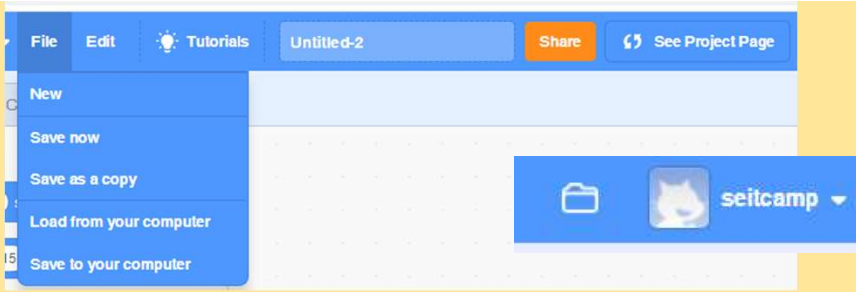
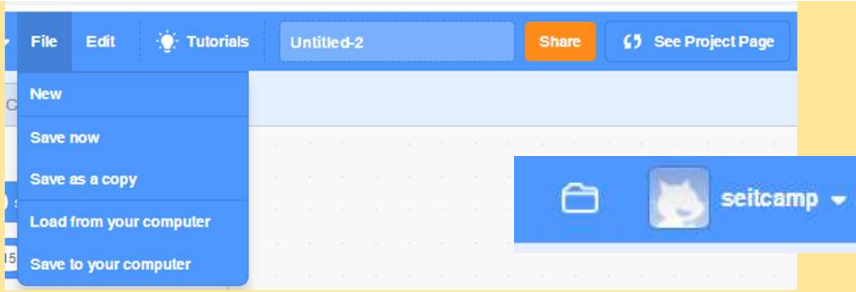
- Check the time!



- **Create a free account to store your code?**
- Workshop-1 http://scratch.mit.edu/signup/3tk2je85j
- (please don't use your real-name)
  - se-cat123, bat345, cat678, se-ironman, se-superman, moew555, …
- Demo: name the file, save, share, remix

- Check the time!



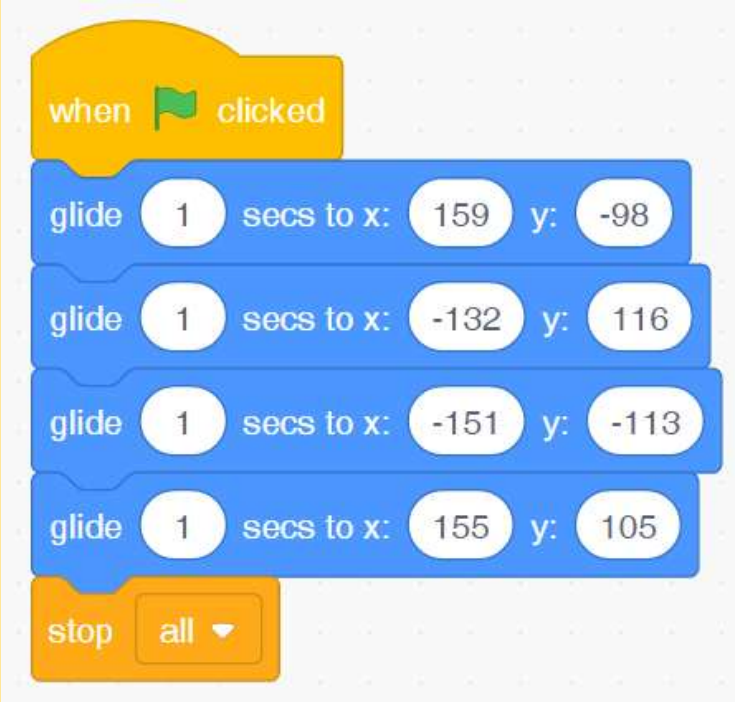
- **Create a free account to store your code?**
- Workshop-2
- http://scratch.mit.edu/signup/fdyxj2yrd
- (please don't use your real-name)
  - se-cat123, bat345, cat678, se-ironman, se-superman, moew555, …

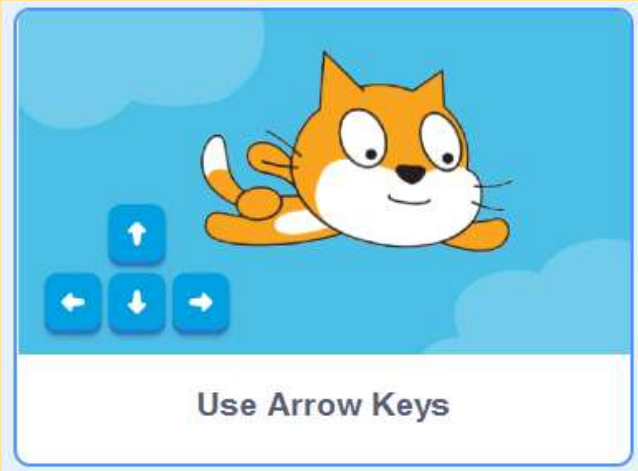
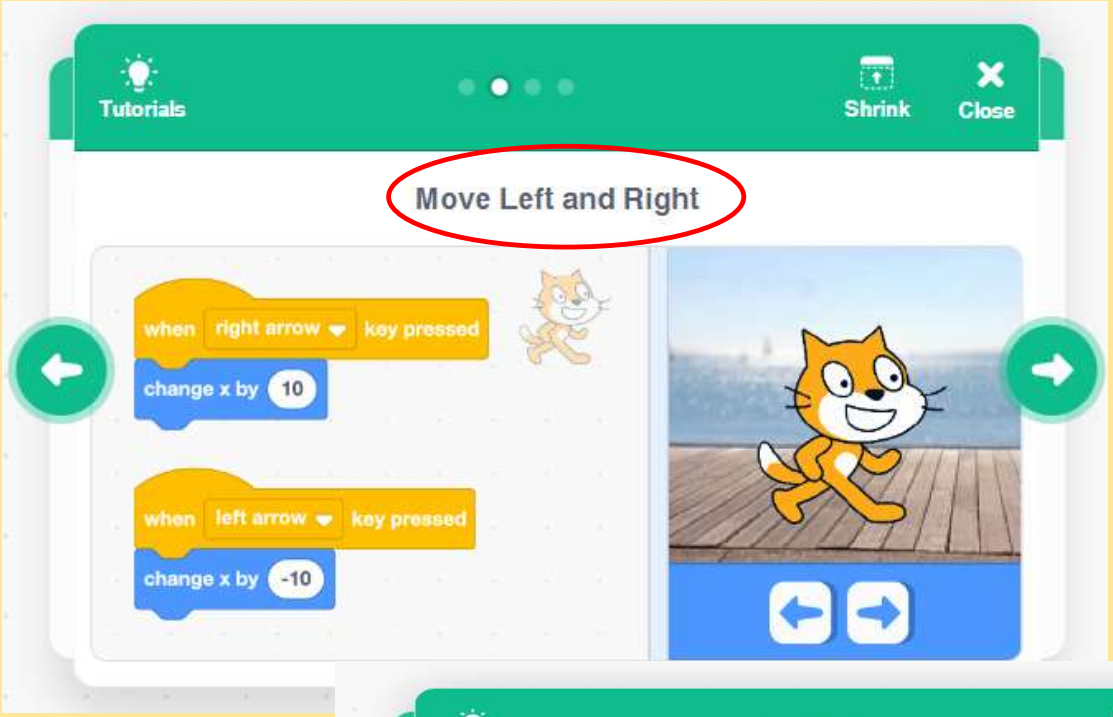
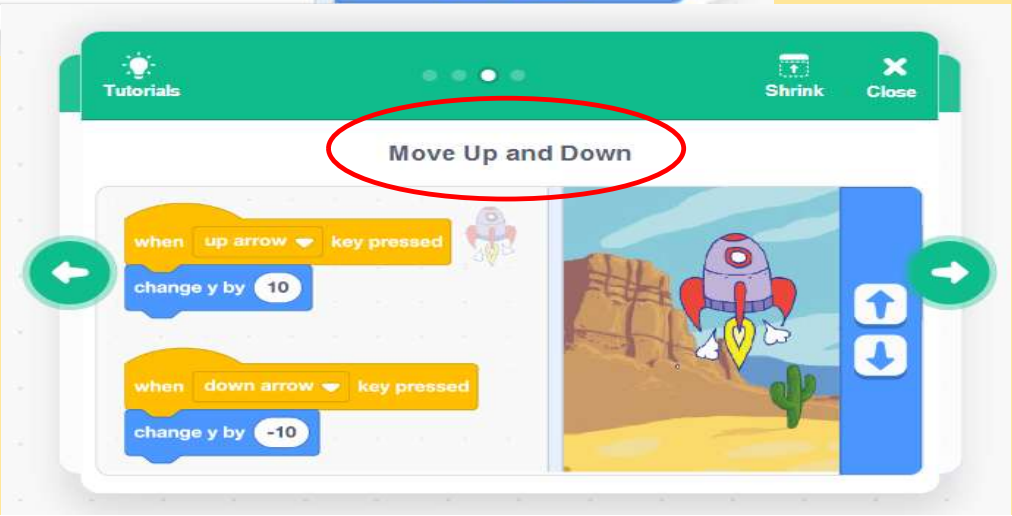
- Demo: name the file, save, share, remix

Challenge: (not homework (:>))

- Can you design a “Pop the Balloon Game for your family?”
- Or Try the “Create A Story” tutorial yourself!

Glide Around





*** Reminder: Watch the video first!*